Indivisible stochastic processes for non-Hermitian and higher-derivative quantum systems: detailed balance, broken $\mathcal{PT}$ symmetry, and the explosion of ghosts

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Abstract

Barandes' stochastic–quantum correspondence represents unitary quantum theory by *indivisible* stochastic processes on a configuration space, while the antilinearity program of Bender and Mannheim replaces Hermiticity by $\mathcal{PT}$ symmetry, recovering unitarity in a modified inner product. We ask whether these two reconstructions are compatible, and follow the question from the canonical 2×2 $\mathcal{PT}$ dimer to interacting Pais–Uhlenbeck-type ghosts. Our results: (i) unbroken antilinear symmetry is exactly the condition for a probability-conserving process to exist *without enlarging the configuration space* (via a time-independent beable map)—on beables generically rotated from the original configurations—and the metric-corrected process is the ordinary (generically indivisible) process of the isospectral Hermitian partner; (ii) a non-Hermitian Hamiltonian admits a *fixed-beable* representation iff its on-site energies are real and its couplings are two-way with positive pair products obeying a Kolmogorov cycle (detailed-balance) condition, and the resulting process is unique—in gauge language, fixed beables exist precisely when the imaginary gauge flux vanishes—the exact obstruction, of which the non-Hermitian skin effect is the spectral manifestation; (iii) the broken phase, treated as an open subsystem by dilation, supports a valid sub-stochastic process whose indivisibility is *terminal* rather than recurrent—we prove exact closed forms, including the generator floor $-(\sin\theta + \tilde{\kappa})$ (a floor on the indivisibility strength $|L_{12}|$, approached from below by the signed entry)—while late-time information backflow vanishes numerically: the two notions of non-Markovianity dissociate; (iv) the exceptional point is a coordinate singularity of the Hermitian dictionary, not of the process; (v) the free Pais–Uhlenbeck ghost admits a perfectly valid fixed-beable process on the truncated configuration space, whereas the no-ghost quantization exits the sample space; (vi) interacting ghosts realize the probabilist's classification conservative/transient/explosive (the last at the mechanism level), and (vii) a deficiency-index/Feller analysis proves an "explosion dictionary" for the one-dimensional monotone mechanism model: one classical integral decides classical completeness, quantum self-adjointness (its limit-circle leg under C^2 error control), and stochastic conservativity, with (viii) a rigorous channel-based multidimensional extension (for separable, central, and boundedly-coupled model classes) in which conservative completions carry an infinite-dimensional, channel-mixing S-matrix at infinity. For the exact quartic Pais–Uhlenbeck operator we further prove an exact displaced-oscillator identity whose channelization (exact up to $O(k^2)$) carries a sign prediction the numerics confirm ($\lambda > 0$: limit-circle channels; $\lambda < 0$: a quantum-completion candidate despite classical blowup), settle deficiency stability at every finite Galerkin channel number, certify

the one-dimensional fiber non-self-adjoint, in a computer-assisted sense, by a glued-tail von Neumann certificate—no compactly supported function can witness deficiency—and conjecturally reduce the remaining $M \to \infty$ step to Gaussian-beam assembly and error control along numerically characterized escape rays. The ghost problem, viewed in the stochastic currency, is a recurrence-classification problem, and the minimal explosive process is its only canonical resolution.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every theorem is stated with a full proof, and every numerical claim is reproducible from the public scripts (see Code availability), where the originating conversation and project briefing are also preserved.

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1 Introduction

Two contemporary research programs demote the standard Hilbert-space formalism of quantum mechanics to a derived structure, and both elevate the role of time reversal.

The first is Barandes' *stochastic–quantum correspondence* [1, 2]: quantum theory is equivalent to a class of generalized stochastic processes on a configuration space, in which a definite configuration (a "beable") exists at all times and the transition matrix $\Gamma(t)$ fails to factorize between measurements. Such processes are *indivisible*: except at special times, no stochastic matrix $\Gamma(t_2, t_1)$ satisfies $\Gamma(t_2) = \Gamma(t_2, t_1)\Gamma(t_1)$. Unitarity, in this reading, is not an axiom but a representation-theoretic consequence.

The second is the antilinearity program of Bender, Mannheim, and collaborators [4, 5, 6]: the correct organizing principle of quantum theory is not Hermiticity but invariance under an antilinear symmetry such as $\mathcal{PT}$. A $\mathcal{PT}$-symmetric, non-Hermitian Hamiltonian with real spectrum and a complete set of symmetry-respecting eigenvectors ("unbroken phase") evolves unitarily with respect to a modified inner product; when eigenvalues form complex-conjugate pairs ("broken phase"), exponentially growing and decaying modes balance in a biorthogonal pairing. The program's flagship application is the fourth-order Pais–Uhlenbeck (PU) oscillator [9], the quantum-mechanical prototype of conformal (Weyl) gravity [7], whose Ostrogradski ghost is claimed to be exorcised by $\mathcal{PT}$-symmetric quantization [8].

These programs have, to our knowledge, never been brought into contact (a recent study of divisibility within the correspondence [3] remains within the standard Hermitian/unitary-generated setting and does not treat non-Hermitian or antilinear dynamics). Barandes' construction rests on ordinary unitarity in a fixed inner product; Mannheim's unitarity lives in a Hamiltonian-dependent inner product. The natural question is:

> *Can antilinear ($\mathcal{PT}$) symmetry predict or produce indivisible stochastic processes—and what becomes of the correspondence in the broken phase, at exceptional points, and for the ghosts it was designed to save?*

This paper answers the question in stages, each stage exactly solvable or numerically verified, and each feeding the next. Our main results:

1. **Existence gates, but does not generate** (Sec. 2). Unbroken antilinear symmetry is precisely the condition for a valid probability-conserving, same-configuration-space process to exist (via a time-independent beable map; generically indivisible); however, the metric-corrected process is the *ordinary* Barandes process of the isospectral Hermitian partner $h = \rho H \rho^{-1}$ [10]. Closed-system $\mathcal{PT}$ machinery is an isomorphism, not a generalization. Its two costs are dynamics-dependent beables and metric non-uniqueness.

2. **Fixed beables $\Leftrightarrow$ detailed balance** (Sec. 3). The *detailed-balance theorem* (Theorem 3.1): a non-Hermitian H admits a beable-preserving (diagonal-metric) representation iff its on-site energies are real and its couplings are two-way with positive pair products satisfying Kolmogorov-type detailed-balance conditions; the metric weights are the inverse stationary distribution, the spectrum is automatically real, and the fixed-beable process is unique—both costs dissolve exactly on this class. In gauge variables the criterion reads: the *imaginary* gauge flux must vanish while real magnetic flux is unconstrained, so the imaginary gauge flux is precisely the obstruction to fixed beables—finer than net spectral winding (Corollary 3.5)—of which the non-Hermitian skin effect [20, 48] is the spectral manifestation (Sec. 3.1).

3. **The broken phase is a valid open process; the two non-Markovianities dissociate** (Sec. 4). Broken-$\mathcal{PT}$ evolution is survival-conditioned dynamics of a contraction; its unitary dilation defines an ordinary indivisible process whose reduction is sub-stochastic. The *terminal-indivisibility theorem* (Theorem 4.1) gives the generator in closed form: divisibility is lost at t_c exactly when $\det\Gamma$ crosses zero, permanently, with floor $-(\sin\theta+\tilde{\kappa})$; the unbroken phase is recurrent instead. Late-window information backflow, by contrast, vanishes (numerically) in the broken phase: Barandes indivisibility is the strictly finer probe.

4. **The exceptional point is a coordinate singularity** (Sec. 5). The metric collapses with Petermann scaling $\mathrm{cond}(\eta)\sim\varepsilon^{-1}$ and the lab-time metric-corrected process freezes to the identity; a nontrivial critical process survives only on the rescaled clock $\tau=\kappa t$, in coordinates whose beable map diverges. The open-system description passes through the exceptional point smoothly.

5. **Pais–Uhlenbeck: the two programs pay in different currencies** (Sec. 6). The Hermitian Ostrogradski realization of free PU is a valid, fixed-beable, indivisible process on a truncated (discretized) configuration space of physical $(x,\dot{x})$—the unbounded-below energy spectrum is invisible to the process. The $\mathcal{PT}$/no-ghost quantization is a domain change whose vacuum has no shadow at any truncation: not a rotation of beables but an exit from the sample space.

6. **Interacting ghosts: conservative, transient, explosive** (Sec. 7). Quadratic interactions collapse the instability threshold from the sum to the difference frequency and make the process *transient* (exact cascade laws), yet it remains a valid doubly stochastic process with balanced energy books. Malicious nonlinear interactions with finite-time classical escape make it *explosive*—at the level of the classical mechanism (its minimal escape process, Sec. 8); the quantum operator can resist (the $\lambda<0$ completion candidates of Sec. 10).

7. **The explosion dictionary** (Sec. 8). The *explosion dictionary* (Theorem 8.1, for the one-dimensional monotone mechanism model; Sec. 10 records where it fails beyond that model): a single classical integral—the time of flight to infinity—decides classical completeness, quantum essential self-adjointness (via the WKB identity "deficiency norm $\approx$ time of flight", its limit-circle leg under C^2 error control), and stochastic conservativity (Feller). Conservative completions reflect probability off infinity at the classical round-trip time; the canonical object is the minimal, explosive, sub-stochastic process.

8. **Multidimensional channel theory** (Sec. 10). Since classical incompleteness and quantum non-self-adjointness can come apart [36], we prove channel-based statements: quadratic ghosts are rigorously safe (Nelson's theorem); separable, central, and boundedly-coupled malicious systems have deficiency indices (∞,∞), so conservative completions carry a channel-mixing *S-matrix at infinity*. The essential self-adjointness of the exact quartic-PU operator is posed as a sharply delimited open problem—and then taken as far as computation allows: an exact displaced-oscillator identity, its channelization (exact up to $O(k^2)$) carrying a confirmed sign prediction (Proposition 10.7), deficiency stability at every finite Galerkin channel number (Theorem 10.8) with its $\lambda<0$ mirror (Corollary 10.9), the no-compact-witness principle (Remark 10.11) together with a computer-assisted glued-tail von Neumann certificate for the one-dimensional fiber, and a Maslov feasibility study con-

jecturally reducing the remaining $M \to \infty$ step to Gaussian-beam assembly and error control along the escape rays.

All numerical claims are reproducible from short standalone scripts (see Code availability).

Result	Sec.	Status	Depends on
A Stochastic–quantum preliminaries ($\Gamma = \lvert U\rvert^2$, divisibility, generator)	§2	R	—
B Existence gate: unbroken antilinearity $\Leftrightarrow$ closed process exists (deflation; two costs)	§2	E+R	A
C Fixed beables $\Leftrightarrow$ Kolmogorov detailed balance (Thm. 3.1)	§3	R	A, B
D Skin effect = the obstruction; topological classification (Cor. 3.5)	§3.1	R	C
E Broken phase: terminal indivisibility (Thm. 4.1); two non-Markovianities dissociate	§4	E+R+N	A, B
F Exceptional point = coordinate singularity of the Hermitian dictionary	§5	E+N	B, E
G Free PU ghost: valid fixed-beable process (truncated configuration space); no-ghost quantization exits the sample space	§6	E+N	A, C
H Interacting ghosts: conservative / transient / explosive	§7	E+N	A, G
I Explosion dictionary, 1D (Thm. 8.1); Feller escape-diffusion lemma	§8	R+N	H
I′ Completions dictionary: self-adjoint extensions $\to$ Feller lateral conditions (Thm. 9.1)	§9	R	I
J Multidimensional channel theory; S-matrix at infinity (model cases, Thm. 10.2)	§10	R	I
J′ Essential self-adjointness of exact quartic PU (Prop. 10.7; glued-tail certificate)	§10	O (N)	J

Status legend: E = exactly solvable; R = rigorous (theorem with full proof, under the hypotheses stated in the cited theorem—including, where present, frame or normalization hypotheses such as the Olver frame); N = numerically verified (reproducible scripts, Code availability); O = open problem, taken as far as computation allows.

Table 1: Results map: each row is a result module, its section, its epistemic status, and the modules it builds on. Arrows run downward: every row depends only on rows above it.

Reading paths. Three routes traverse the map. The *foundations reader* ($\mathcal{PT}$/antilinearity and the correspondence) should take §2–§5 in order (A–F): gate, detailed balance, broken phase, exceptional point. The *skin-effect reader* needs only §2 and then §3 (C–D), where the imaginary gauge flux is identified as the exact obstruction to fixed beables. The *ghost/self-adjointness reader* may proceed directly from the preliminaries (§2) to §6–§10 (G–J′), which are independent of the broken-phase and exceptional-point material and need from §3 only the statement of Theorem 3.1 (trivial in the Hermitian case); the analytic core is the explosion dictionary of §8 and its completions dictionary (§9). Experimental signatures for all three routes are collected in §11.

Part I
The correspondence machinery

2 Preliminaries and the deflationary observation

2.1 The stochastic–quantum correspondence

Following [1, 2], a *generalized stochastic system* on a configuration set $\{1, \dots, n\}$ is specified by a family of column-stochastic matrices $\Gamma(t)$, $\Gamma_{ij}(t) \geq 0$, $\sum_i \Gamma_{ij}(t) = 1$, giving the probability of configuration i at time t given j at time 0. The system is *divisible* on (t_1, t_2) if there is a stochastic matrix $\Gamma(t_2, t_1)$ with $\Gamma(t_2) = \Gamma(t_2, t_1)\,\Gamma(t_1)$, and *indivisible* if such factorizations fail except at isolated times ("division events"). The stochastic–quantum dictionary represents

$$\Gamma(t) = \Theta(t) \odot \Theta^*(t), \qquad \Theta(t) = U(t) = \mathrm{e}^{-\mathrm{i}Ht}, \qquad \Gamma_{ij} = |U_{ij}|^2, \tag{1}$$

with $\odot$ the entrywise product; $|U|^2$ is doubly stochastic if U is unitary (the converse fails: stochasticity of $|U|^2$ fixes only the moduli), and indivisibility expresses interference, $|UV|^2 \neq |U|^2|V|^2$. Two facts matter below: the family $\Gamma(t)$ need not compose—a *necessary* signature of indivisibility, whose operative test is the negativity of the divisor (2), not the mere size of $\Gamma(2t) - \Gamma(t)^2$—and only per-time unistochasticity is required.

If $\Gamma(t_1)$ is invertible, the candidate divisor is unique,

$$M = \Gamma(t_2)\,\Gamma(t_1)^{-1}, \tag{2}$$

and has unit column sums automatically; divisibility on (t_1, t_2) holds iff M is entrywise nonnegative (with column sums ≤ 1 in the sub-stochastic case). Infinitesimally, with the generator $L(t) = \dot{\Gamma}(t)\Gamma(t)^{-1}$, divisibility across t (classical P-divisibility) holds iff the off-diagonal entries of $L(t)$ are nonnegative (with column sums of $L(t)$ nonpositive in the sub-stochastic case).

2.2 Pseudo-Hermitian and $\mathcal{PT}$-symmetric quantum mechanics

A diagonalizable H with antilinear symmetry has eigenvalues either real or in complex-conjugate pairs [5, 6]. If the spectrum is real and a positive-definite metric η exists with

$$H^\dagger \eta = \eta H, \tag{3}$$

then $\rho = \eta^{1/2}$ furnishes a Hermitian partner $h = \rho H \rho^{-1}$ and a unitary $u(t) = \rho\,\mathrm{e}^{-\mathrm{i}Ht}\rho^{-1}$ [10]. The canonical biorthogonal choice is $\eta = (VV^\dagger)^{-1}$, V the right-eigenvector matrix. Our laboratory is the canonical dimer

$$H(\theta, s) = \begin{pmatrix} \mathrm{e}^{\mathrm{i}\theta} & s \\ s & \mathrm{e}^{-\mathrm{i}\theta} \end{pmatrix}, \qquad E_\pm = \cos\theta \pm \sqrt{s^2 - \sin^2\theta}, \tag{4}$$

which is $\mathcal{PT}$-symmetric ($PH^*P = H$, $P = \sigma_x$): unbroken for $s > \sin\theta$, broken for $s < \sin\theta$, with an exceptional point (EP) at $s_* = \sin\theta$.

2.3 Existence gates; deflation; two costs

Throughout we use $\theta = \pi/4$ and, unless stated, $t = 1.3$.

Proposition 2.1 (metric-corrected process; deflation)**.** *In the unbroken phase,* $\Gamma(t) = |\rho\, \mathrm{e}^{-\mathrm{i}Ht}\rho^{-1}|^2$ *is doubly stochastic and, at generic tested parameters, indivisible; in the broken phase no positive-definite solution of* (3) *exists, and—for any fixed invertible beable map—no probability-conserving process of this form exists (time-dependent identifications, as on the rescaled clock of Sec. 5, are a separate question). Moreover* $\rho\, \mathrm{e}^{-\mathrm{i}Ht}\rho^{-1} = \mathrm{e}^{-\mathrm{i}ht}$*: the metric-corrected process is the ordinary Barandes process of the Hermitian partner* h*.*

Proof. If $\eta = \rho^2$ solves (3) with ρ positive definite, then $h := \rho H \rho^{-1}$ obeys $h^\dagger = \rho^{-1}H^\dagger\rho = \rho^{-1}(\eta H \eta^{-1})\rho = h$, so h is Hermitian, $\rho\, \mathrm{e}^{-\mathrm{i}Ht}\rho^{-1} = \mathrm{e}^{-\mathrm{i}ht}$ is unitary, and the entrywise modulus-squared of a unitary are doubly stochastic (unistochastic). Indivisibility at generic t is exhibited numerically below. In the broken phase the spectrum contains a complex-conjugate pair; similarity to a Hermitian h would force a real spectrum, so no positive-definite solution of (3)—and hence no process of this form—exists. The exclusion is in fact similarity-free: for any fixed invertible B, column-stochasticity of $|B\mathrm{e}^{-\mathrm{i}Ht}B^{-1}|^2$ makes $\left\|B\mathrm{e}^{-\mathrm{i}Ht}B^{-1}\right\| = \sqrt{n}$ (Frobenius) bounded in t, yet the Frobenius norm dominates the spectral radius, which is similarity-invariant: $\sqrt{n} \geq \mathrm{spr}(B\mathrm{e}^{-\mathrm{i}Ht}B^{-1}) = \mathrm{spr}(\mathrm{e}^{-\mathrm{i}Ht}) = \mathrm{e}^{\beta t}$, $\beta = \max_k \mathrm{Im}\, E_k > 0$ in the broken phase—impossible for large t. Hence no invertible B yields a probability-conserving process on any beables. The same constraint closes the boundary case $s = s_*$: column-stochasticity forces $\left\|B\mathrm{e}^{-\mathrm{i}Ht}B^{-1}\right\| = \sqrt{n}$ at *every* t, so $\{\mathrm{e}^{-\mathrm{i}Ht}\}_{t\geq 0}$ is similar, via the fixed B, to a bounded family—impossible for a defective H, whose Jordan block makes $\left\|\mathrm{e}^{-\mathrm{i}Ht}\right\|$ grow linearly in t. Real *diagonalizable* spectrum is thus necessary, covering the exceptional point as well. □

Numerically, at $s = 1$ the naive column sums of $|U|^2$ are $(3.23, 1.30)$ (here and throughout, $\|\cdot\|$ on matrices is the Frobenius norm), while the corrected process is $\Gamma = \left(\begin{smallmatrix} 0.368 & 0.632 \\ 0.632 & 0.368 \end{smallmatrix}\right)$, indivisible at this t: the divisor (2) has a negative entry, $\min M = -1.13$ (the family also fails to compose, $\left\|\Gamma(2t) - \Gamma(t)^2\right\| \approx 0.93$); at $s = 0.5$ (spectrum $0.7071 \pm 0.5\mathrm{i}$) every Hermitian intertwiner is indefinite and $\left\|\mathrm{e}^{-\mathrm{i}Ht}\right\|$ grows as 1.58, 2.04, 3.61, 10.4, 77.2 at $t = 0.5, 1, 2, 4, 8$. Thus unbroken antilinearity is exactly the *gatekeeper* for process existence, while the process itself is isomorphic to an ordinary one—closed-system $\mathcal{PT}$ machinery cannot produce a new class of indivisible processes. The isomorphism carries two costs: (a) ρ is generically non-diagonal, so the beables rotate, and the rotation depends on H; (b) the metric, hence the process, is non-unique. Both costs are resolved exactly in the next section.

3 Fixed beables and Kolmogorov detailed balance

We ask which H admit a metric diagonal in the configuration basis, so that the beables are preserved.

Theorem 3.1 (fixed-beable characterization)**.** *A matrix* H *admits a positive diagonal metric* $\eta = \mathrm{diag}(d)$*,* $d_i > 0$*, satisfying* (3) *iff all of the following hold:*

(K1) *every diagonal entry* H_{ii} *is real;*

(K2) $H_{ij} = 0 \Leftrightarrow H_{ji} = 0$*;*

(K3) $H_{ij}H_{ji} > 0$ *for every coupled pair* $i \neq j$*;*

(K4) (Kolmogorov cycle condition) *around every cycle of the coupling graph, the product of the moduli $|H_{ij}|$ taken one way equals the product taken the other way.*

When they hold, d is unique up to one positive scale per connected component of the coupling graph.

Proof. The component form of (3) with $\eta = \mathrm{diag}(d)$ is $d_i H_{ij} = d_j \overline{H_{ji}}$. The $i = j$ component gives (K1); positivity of d forces (K2); the phase of the off-diagonal relation gives (K3); its modulus gives $d_i/d_j = |H_{ji}|/|H_{ij}|$, whose consistency around closed loops is (K4). Conversely, define d along a spanning forest; (K4) makes it well defined, and positivity is automatic. □

Corollary 3.2. *Under* (K1)–(K4)*: (i) $h = \eta^{1/2} H \eta^{-1/2}$ is Hermitian, so the spectrum of H is automatically real—unbroken antilinearity is derived, not assumed; (ii) with $\rho = \mathrm{diag}(\sqrt{d})$ the process is the pure reweighting $\Gamma_{ij}(t) = (d_i/d_j)\,|U_{ij}(t)|^2$, invariant under the residual scale freedom, hence* unique*; (iii) with classical hopping rates $q(i{\to}j) := |H_{ji}|$, the distribution $\pi_i \propto 1/d_i$ satisfies detailed balance $\pi_i q(i{\to}j) = \pi_j q(j{\to}i)$: the metric weights are the inverse Kolmogorov equilibrium distribution [17, 18]; (iv) the fixed-beable class is exactly the positive-diagonal similarity orbit of the Hermitian matrices ("detailed-balanced gain/loss").*

(A biorthogonal-KMS route to the same metric-versus-detailed-balance equivalence, for diagonalizable non-Hermitian H of real spectrum, was given independently by Lan, Ma and Yang [61]; the fixed-beable/Kolmogorov reading, the inverse-stationary weights, and the skin-effect obstruction below are ours.)

Three instructive cases (all verified numerically, including 200/200 randomized trials at $n = 3, \ldots, 6$ in which an independent null-space computation confirms the solution space is exactly the constructive d's ray):

(a) $n = 2$. Cycles beyond pairs are absent, so (K4) is vacuous: fixed-beable $\Leftrightarrow H_{11}, H_{22} \in \mathbb{R}$ and $H_{12}H_{21} > 0$ (or $H_{12} = H_{21} = 0$, the decoupled case). The dimer (4) fails (K1) for every $\theta \neq 0, \pi$: within that family, fixed beables $\Leftrightarrow$ Hermiticity, consistent with the rotation found in Sec. 2. The cycle condition first bites at $n \geq 3$.

(b) A genuinely non-Hermitian fixed-beable 3×3 example. Take $d = (1, 2, 4)$, real diagonal $(0.3, -0.4, 1.0)$, upper entries $H_{12} = 1 + \mathrm{i}$, $H_{13} = 0.8\mathrm{i}$, $H_{23} = 1.5\,\mathrm{e}^{\mathrm{i}\pi/5}$ and lower entries $H_{ji} = (d_i/d_j)\overline{H_{ij}}$. Then $\|H - H^\dagger\| = 1.69$, yet the spectrum is real $\{-1.319,\ 0.210,\ 2.009\}$; the naive column sums $(0.40, 1.17, 3.04)$ are repaired to a doubly stochastic Γ by the reweighting alone—no beable rotation—and the resulting process is indivisible (divisor (2) minimum entry -0.26; the family does not compose, $\|\Gamma(2t) - \Gamma(t)^2\| \approx 0.41$).

(c) Only (K4) fails. $H = \mathrm{diag}(1, 2, 3)$ plus the asymmetric positive ring with $(H_{12}, H_{23}, H_{31}) = (0.6, 0.6, 0.6)$ and $(H_{21}, H_{32}, H_{13}) = (0.3, 0.3, 0.3)$: cycle products $0.216 \neq 0.027$. The spectrum is still real $\{0.847,\ 1.840,\ 3.313\}$, a positive-definite (non-diagonal) metric exists and yields a valid indivisible process, but the diagonal-intertwiner solution space is exactly zero and ρ has an 18% off-diagonal fraction: *(K4) gates fixed beables specifically, not process existence.*

Remark 3.3 (relation to prior work). The matrix-algebra core of Theorem 3.1 is classical: diagonal symmetrizability of nonnegative matrices was characterized by Parter and Youngs [37], and cyclic-product criteria for diagonal similarity by Engel and Schneider [38]. The contribution here is the identification of this class with *beable preservation* in the stochastic–quantum correspondence, the detailed-balance dictionary for the metric, the automatic reality of the spectrum in the pseudo-Hermitian setting, and the uniqueness of the resulting process. In the reverse direction—non-Hermitian quantum techniques applied to classical Markov processes violating detailed balance—see [39].

3.1 Gauge form: the non-Hermitian skin effect is the obstruction

Conditions (K2)–(K3) have a gauge-theoretic meaning: they hold iff the couplings can be written *uniquely* as

$$H_{jk} = t_{jk}\, \mathrm{e}^{\mathrm{i}\varphi_{jk}+w_{jk}}, \qquad t_{jk} = t_{kj} > 0, \quad \varphi_{jk} = -\varphi_{kj}, \quad w_{jk} = -w_{kj}, \tag{5}$$

i.e. as a lattice Peierls substitution $p \to p - A - \mathrm{i}W$ with a *real* gauge field φ (link variables of A) and an *imaginary* gauge field w (link variables of W, the Hatano–Nelson field [20]); (K1) is the absence of on-site gain/loss. In these variables the detailed-balance theorem (Theorem 3.1) becomes:

Theorem 3.4 (skin obstruction)**.** *For a gauge-form Hamiltonian (5) with real on-site potential:*

(i) a fixed-beable representation exists iff the imaginary flux vanishes through every cycle, $\sum_C w = 0$*; then $w_{jk} = \chi_j - \chi_k$ is pure gauge, the metric is the exponentiated imaginary gauge transformation $\eta = \mathrm{diag}(\mathrm{e}^{-2\chi})$, and $\Gamma_{jk}(t) = \mathrm{e}^{2(\chi_k-\chi_j)}|U_{jk}(t)|^2$. The* real *flux (magnetic field) is unconstrained.*

(ii) With classical rates $q(j\to k) = |H_{kj}|$, the imaginary flux is, up to the cycle-orientation convention, half the cycle affinity of the rate graph [19]: it vanishes iff the classical chain is reversible (Kolmogorov), and nonzero imaginary flux is equivalent to a nonzero steady-state cycle current [19].

(iii) (Hatano–Nelson) A ring ($N \geq 3$) with uniform $w \neq 0$ carries flux Nw: no fixed beables exist. For the uniform *ring (uniform t, uniform w, zero real hopping phase, constant on-site potential) the periodic-boundary spectrum is moreover a complex loop (the spectral winding that classifies the skin effect [48])—explicitly $E_k = V_0 + 2t\cos(k - \mathrm{i}w)$, $k \in \frac{2\pi}{N}\mathbb{Z}$, non-real for every $N \geq 3$ since $\mathrm{Im}\, E_k = 2t \sinh w \sin k \neq 0$ for $k \neq 0, \pi$—so by Proposition 2.1 no closed process exists there at all; a non-uniform ring with nonzero net flux still has no fixed beables, but its flux can be too weak to break spectral reality, leaving a closed rotated-beable process possible (the diagonally dominated triangle below is exactly such a case). An open chain is a tree: fixed beables exist at every finite N with weights $d_i = \mathrm{e}^{2wi}$; the boundary-condition dichotomy of the skin effect is the flux/pure-gauge dichotomy.*

Proof. (i) restates (K4) in the variables (5): cycle products of moduli agree in the two orientations iff $\mathrm{e}^{\sum_C w} = \mathrm{e}^{-\sum_C w}$. (ii) is Kolmogorov's criterion applied to q, whose cycle ratios are $\mathrm{e}^{2\sum_C w}$. (iii) a ring has one independent cycle; a tree has none. □

Numerically (Fig. 1): over 50 random gauge-form graphs with random *real* fluxes and pure-gauge w, the constructed metric is $\mathrm{e}^{-2\chi}$ exactly and the spectrum real; injecting imaginary flux through any one cycle collapses the diagonal-intertwiner space to zero. For the Hatano–Nelson chain the OBC skin eigenvectors equal the gauge factor e^{-wi} times the Hermitian-chain eigenvectors to 5×10^{-15}, and $\mathrm{cond}(\eta) = \mathrm{e}^{2w(N-1)}$ (an analytic identity, verified to four digits)—so the fixed-beable representation, while exact at every finite N, degenerates exponentially in the thermodynamic limit, reproducing the unbounded-metric pathology of Sec. 6 in one-dimensional lattice form. On the classical side, the steady-state ring current vanishes (to 10^{-17}) exactly when the imaginary flux does. Finally, in a regime where the flux is too weak to break spectral reality (a diagonally dominated triangle), the binary criterion trips at any nonzero flux while the

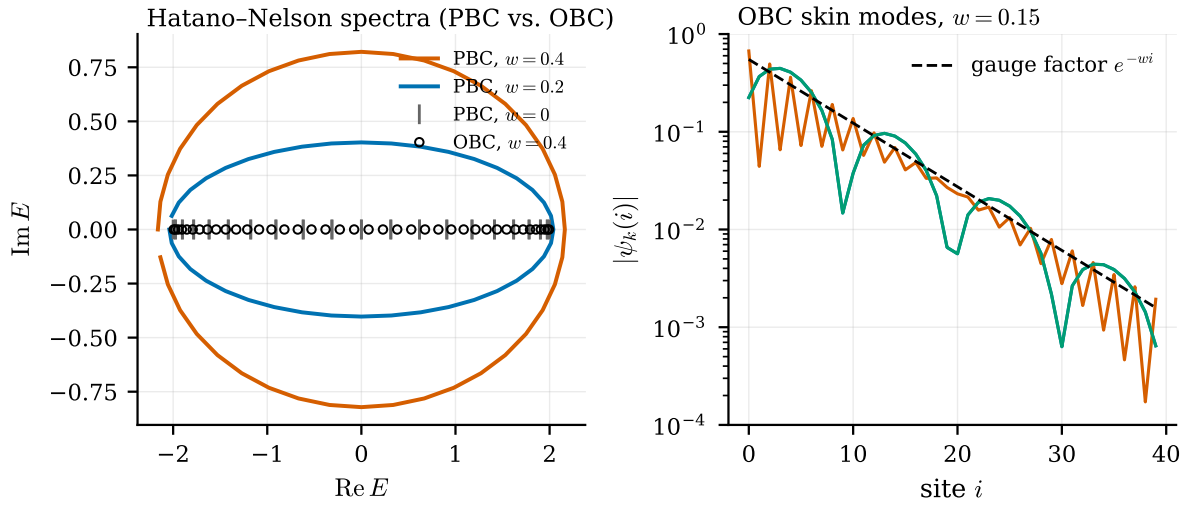


Figure 1: The skin-obstruction theorem (Theorem 3.4). Left: Hatano–Nelson spectra. Nonzero imaginary flux (periodic boundaries, $w \neq 0$) winds the spectrum into a complex loop—no closed process, no fixed beables; the flux-free cases ($w = 0$, or any w with open boundaries) have real spectra. Right: open-boundary skin modes $|\psi_k(i)|$ ride the pure-gauge envelope e^{-wi} (dashed): the skin pileup is precisely the beable-preserving metric $d_i = \mathrm{e}^{2wi}$ in disguise, exact at finite N and exponentially degenerate as $N \to \infty$.

beable rotation grows continuously with it: the obstruction is quantitative below the antilinear transition.

The non-Hermitian skin effect—spectral winding under periodic boundaries, boundary pileup under open ones—is thus the spectral and boundary face of the obstruction to a beable-preserving stochastic representation, whose exact invariant is the imaginary gauge flux (the H^1 class of Corollary 3.5, finer than net winding); the obstruction equals detailed-balance violation (irreversibility) of the underlying classical rate graph.

Corollary 3.5 (topological classification of the obstruction)**.** *For gauge-form couplings on a graph or lattice, the fixed-beable obstruction is the class of the imaginary gauge field in the first cohomology $H^1(G, \mathbb{R})$: the imaginary* field strength *must vanish through every contractible cycle (no imaginary magnetic flux), and the imaginary* holonomy *through every non-contractible one (no winding)—on a d-torus, an $\mathbb{R}^d$ of independent windings, matching the multi-directional classification of skin effects [48], plus the pointwise plaquette condition that band-language winding does not see. The obstruction is finer than net spectral winding: two Hatano–Nelson chains with* opposite *fields (net winding zero—the doubled structure of the $\mathbb{Z}_2$ skin effect [48]), coupled by reciprocal rungs, are obstructed through their inter-chain cycles of circulation $2w$.*

Numerically (`scripts/skin_topology_beables.py`): on a 3×4 torus the diagonal-metric solution space has dimension 1 iff both windings and the plaquette flux vanish, and 0 if any is switched on; on a cylinder, "winding" along the open direction is gauge-removable while the periodic direction obstructs—the obstruction lives in H^1 of the actual geometry; and the $\mathbb{Z}_2$-type ladder has solution-space dimension 2 uncoupled (one metric per chain, $\mathrm{e}^{\pm 2wi}$) collapsing to 0 at any rung coupling despite zero net winding.

4 The broken phase as an open subsystem

4.1 Broken $\mathcal{PT}$ dynamics is survival-conditioned dynamics

Write (4) as the balanced gain/loss form $H = \cos\theta\,\mathbf{1} + \mathrm{i}\sin\theta\,\sigma_z + s\,\sigma_x$. Subtracting the uniform gain,

$$H_{\mathrm{loss}} = H - \mathrm{i}\sin\theta\,\mathbf{1}, \qquad \mathrm{e}^{-\mathrm{i}Ht} = \mathrm{e}^{\sin\theta\,t}\,C(t), \quad C(t) = \mathrm{e}^{-\mathrm{i}H_{\mathrm{loss}}t}, \tag{6}$$

where $C(t)$ is a contraction (singular values ≤ 1; verified to machine precision). The notorious broken-phase runaway is thus a scalar normalization: broken-$\mathcal{PT}$ evolution *is* the no-leak-conditioned dynamics of a lossy system.

For each t the Halmos dilation [12, 13]

$$U_{\mathrm{big}}(t) = \begin{pmatrix} C(t) & \sqrt{\mathbf{1} - C(t)C(t)^\dagger} \\ \sqrt{\mathbf{1} - C(t)^\dagger C(t)} & -C(t)^\dagger \end{pmatrix} \tag{7}$$

is exactly unitary with $\Gamma_{\mathrm{big}}(0) = \mathbf{1}_4$. Because the correspondence requires only per-time unistochasticity—the family need not compose—$\Gamma_{\mathrm{big}}(t) = |U_{\mathrm{big}}(t)|^2$ is a Barandes *indivisible* process (per-time unistochastic, with a negative divisor entry, not one-parameter-group-generated) on four configurations $\{1, 2, a_1, a_2\}$ (two system sites, two "leaked" ancilla configurations), and the reduction

$$\Gamma_{\mathrm{red}}(t) = |C(t)|^2 \tag{8}$$

is a sub-stochastic process (column deficits = leak probabilities), valid at all times in the phase where the closed construction failed totally. (A unitary-embedding perspective on $\mathcal{PT}$ dynamics—a "hidden entangled partner" restoring lost information—was developed by Kawabata, Ashida, and Ueda [40]; the per-time Halmos dilation used here is its process-level counterpart. An operator-theoretic dilation of *indivisible* contraction families to divisible/unitary ones on graphs was given independently by Dahya [60], without the $\mathcal{PT}$/beable or Barandes-process reading developed here.)

4.2 Exact generator: terminal indivisibility

Theorem 4.1 (terminal indivisibility in the broken phase)**.** *Let* $0 < s < \sin\theta$, $\tilde{\kappa} = \sqrt{\sin^2\theta - s^2} > 0$, *and set (notation local to this theorem and its proof)* $c = \cosh\tilde{\kappa}t$, $\sigma = \sinh(\tilde{\kappa}t)/\tilde{\kappa}$,

$$p = c + \sin\theta\,\sigma, \qquad q = c - \sin\theta\,\sigma, \qquad r = s\,\sigma.$$

Then

$$\Gamma_{\mathrm{red}}(t) = \mathrm{e}^{-2\sin\theta\,t}\begin{pmatrix} p^2 & r^2 \\ r^2 & q^2 \end{pmatrix}, \qquad \det\Gamma_{\mathrm{red}} = \mathrm{e}^{-4\sin\theta\,t}\,(1 - 2r^2), \tag{9}$$

and the generator $L = \dot{\Gamma}_{\mathrm{red}}\Gamma_{\mathrm{red}}^{-1}$ *has the exact off-diagonal entries*

$$L_{12} = \frac{2spr}{1 - 2r^2}, \qquad L_{21} = \frac{2sqr}{1 - 2r^2}. \tag{10}$$

Consequently:

(i) the process is P-divisible precisely on $[0, t_c)$ *with* $t_c = \tilde{\kappa}^{-1}\,\mathrm{arcsinh}\big(\tilde{\kappa}/(s\sqrt{2})\big)$, *which is exactly the time at which* $\det\Gamma_{\mathrm{red}}$ *crosses zero;*

(ii) $L_{12} < 0$ *strictly for* all $t > t_c$*: permanent (terminal) indivisibility;*

(iii) $L_{12} < -(\sin\theta + \tilde{\kappa})$ *for all* $t > t_c$*, with* $L_{12} \to -(\sin\theta + \tilde{\kappa})$ *as* $t \to \infty$*: the limit is a floor on* $|L_{12}|$*, approached from below by the signed entry;*

(iv) L_{21} *returns positive at* $t_0 = \tilde{\kappa}^{-1} \operatorname{artanh}(\tilde{\kappa}/\sin\theta) > t_c$*: late-time indivisibility is carried entirely by transitions into the gain site;*

(v) the same formulas continue to the unbroken phase ($\tilde{\kappa} \to \mathrm{i}\kappa$*, i.e.* $\kappa = \sqrt{s^2 - \sin^2\theta}$*), where* $1 - 2r^2 = 1 - 2(s/\kappa)^2 \sin^2 \kappa t$ *with* $(s/\kappa)^2 > 1$ *oscillates through zero forever (recurrent indivisibility windows of period* π/κ*), and to the exceptional point, where everything is polynomial with* $r = st$ *and* $t_c = 1/(s\sqrt{2})$*, continuously in* s*.*

Proof. Since $A = \mathrm{i}\sin\theta\,\sigma_z + s\sigma_x$ obeys $A^2 = (s^2 - \sin^2\theta)\mathbf{1} = -\tilde{\kappa}^2\mathbf{1}$, $\mathrm{e}^{-\mathrm{i}At} = c\,\mathbf{1} - \mathrm{i}\sigma A$, giving (9) entrywise. With $r' = sc$ and $p' = \tilde{\kappa}^2\sigma + \sin\theta\, c$ one finds the Wronskian-type identities

$$r'p - p'r \;=\; r'q - q'r \;=\; s\,(c^2 - \tilde{\kappa}^2\sigma^2) \;=\; s,$$

constants in t; inserting them into $L = \dot{\tilde{\Gamma}}\tilde{\Gamma}^{-1} - 2\sin\theta\,\mathbf{1}$ (the scalar factor shifts only the diagonal) yields (10); the reduced divisor stays sub-stochastic, $\mathbf{1}^T L = (0, -4\sin\theta) \le 0$ for all t. Statements (i)–(v) follow by inspection of signs: $p, r > 0$ always, $1 - 2r^2$ crosses zero once (hyperbolic) at t_c, q crosses zero once at $t_0 > t_c$; for (iii), $2spr + (\sin\theta + \tilde{\kappa})(1 - 2r^2) = 2s^2\sigma(c - \tilde{\kappa}\sigma) + \sin\theta + \tilde{\kappa} = 2s^2\sigma\,\mathrm{e}^{-\tilde{\kappa}t} + \sin\theta + \tilde{\kappa} > 0$ while $1 - 2r^2 < 0$ for $t > t_c$, so $L_{12} < -(\sin\theta + \tilde{\kappa})$ there, and $L_{12} \to -(\sin\theta + \tilde{\kappa})$ from the asymptotics $c \sim \tilde{\kappa}\sigma \sim \frac{1}{2}\mathrm{e}^{\tilde{\kappa}t}$. □

The predictions are exact against brute-force numerics (Fig. 2; closed forms to 10^{-14}, generator to 10^{-11}; the observed floor -1.207 at $s = 0.5$ is $-(\sin\frac{\pi}{4} + 0.5) = -1.2071$; the observed first indivisible time 1.35 on a 0.05 grid is $t_c = 1.3170$).

4.3 The two non-Markovianities dissociate

For open systems, non-Markovianity is commonly diagnosed by information backflow (increase of trace/total-variation distance between two evolutions) [14, 15, 16]. On the classical marginal with the leak recorded as a third outcome, the total-variation distance is numerically monotone in *both* phases (no classical backflow observed, though with divisibility lost after t_c monotonicity is no longer guaranteed); on the survival-conditioned (post-selected) process, the late-window backflow vanishes for every broken s whose indivisibility onset $t_c(s)$ precedes the window (all $s \gtrsim 10^{-6}$ here; conditional backflow in the broken phase is confined to a neighborhood of $t_c(s)$, which grows as $\tilde{\kappa}^{-1}\ln(\sqrt{2}\,\tilde{\kappa}/s)$ only as $s \to 0$) and jumps at the EP (e.g. $0.000 \to 1.32$ across $s_* = 0.7071$ at $s = 0.72$, window $t \in [20, 40]$)—the stochastic-level analogue of the reversible–irreversible criticality of information flow established for $\mathcal{PT}$ systems by Kawabata, Ashida, and Ueda [40]. Meanwhile, Barandes indivisibility is present in *both* phases; what changes at the EP is its temporal structure, recurrent ↔ terminal (the terminal-indivisibility theorem, Theorem 4.1). Figure 3 contrasts the two diagnostics across the transition: the backflow order parameter switches at the EP, while the indivisibility onset $t_c(s)$ is finite and *continuous* through it.

Thus the broken phase is *backflow-Markovian in the late window yet permanently Barandes-indivisible*: the two notions genuinely dissociate, indivisibility being strictly finer. That the relation between divisibility- and information-flow-based notions of Markovianity is subtle for

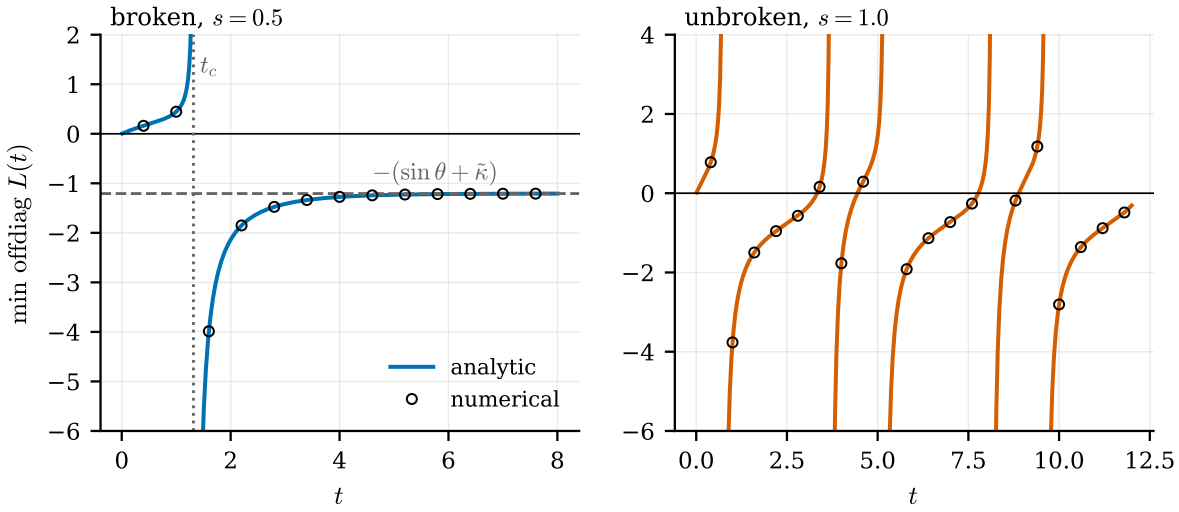


Figure 2: Minimum off-diagonal entry of the generator $L(t) = \dot{\Gamma}_{\mathrm{red}}\Gamma_{\mathrm{red}}^{-1}$ of the reduced process (8): analytic closed forms (10) (curves) versus brute-force numerics (circles). Left: broken phase ($s = 0.5$)—divisibility is lost at t_c, exactly where $\det\Gamma_{\mathrm{red}}$ crosses zero, and never recovers; the generator approaches the strictly negative floor $-(\sin\theta + \tilde{\kappa})$ (Theorem 4.1): *terminal* indivisibility. Right: unbroken phase ($s = 1.0$)—the same closed forms continued to trigonometric functions give *recurrent* windows of divisibility and indivisibility at the Rabi period (curves interrupted where $\det\Gamma_{\mathrm{red}}$ passes through zero).

quantum channels—equivalent for invertible maps [41], genuinely inequivalent for noninvertible ones [42]—is known; the $\mathcal{PT}$ dimer makes the dissociation exact, phase-locked, and—in the broken phase—permanent, at the level of the classical configuration process. A closed Hermitian system, being quasi-periodic, can only be recurrently indivisible; terminal indivisibility with monotone leak is the stochastic signature of the broken phase. (At $n = 2$ every lossy two-site Hamiltonian *with degenerate site frequencies and reciprocal coupling* is a shifted $\mathcal{PT}$ dimer, so there "overdamped open dynamics" and "broken $\mathcal{PT}$" coincide; a detuned pair adds a real σ_z component the $\mathcal{PT}$ form does not carry.)

5 The exceptional point is a coordinate singularity

At $s_* = \sin\theta$ the dimer is defective (a Jordan block; numerically $\mathrm{rank}(H - \cos\theta\,\mathbf{1}) = 1$), the naive process grows polynomially ($|U|^2$ column sums $\sim t^2$; fitted exponent 1.95), and the two-dimensional space of Hermitian intertwiners consists of one *singular positive-semidefinite rank-one ray* and one indefinite direction: the positive-definite cone is exited exactly through its boundary, and ρ does not exist.

Approaching from the unbroken side, $s = s_*(1 + \varepsilon)$, $\kappa \simeq \sin\theta\sqrt{2\varepsilon}$, we find clean power laws (Fig. 4; log–log fits over $\varepsilon = 10^{-1} \ldots 10^{-7}$, all within ± 0.0025):

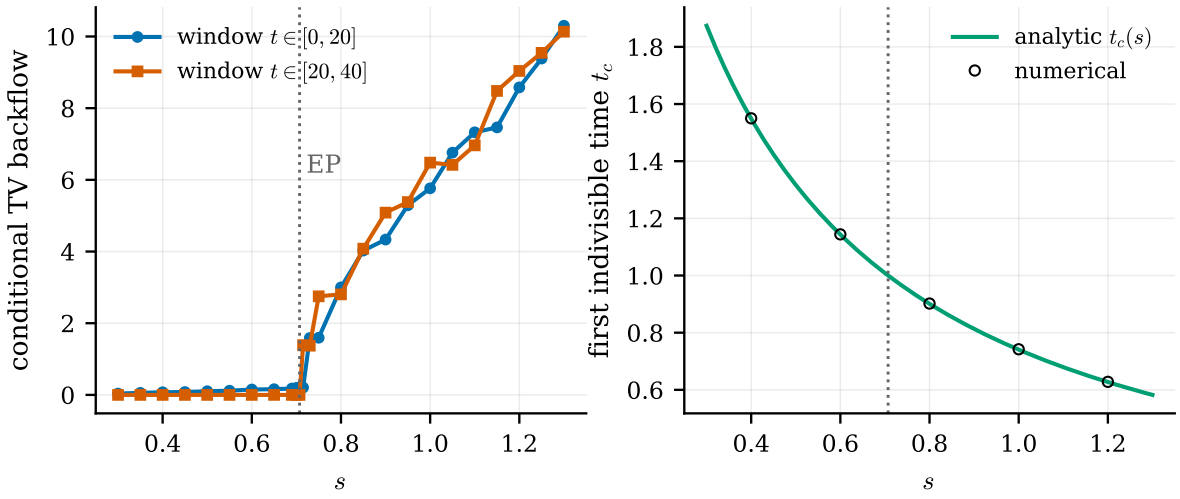


Figure 3: The two non-Markovianities across the $\mathcal{PT}$ transition (EP at $s_* = \sin\frac{\pi}{4} \approx 0.707$, dotted line). Left: conditional total-variation backflow in an early ($t \in [0, 20]$) and a late ($t \in [20, 40]$) window; the late-window backflow vanishes in the broken phase (for onsets $t_c(s)$ preceding the window) and switches on at the EP (the apparent smoothness just above s_* is the diverging Rabi period π/κ meeting the finite window). Right: the first indivisible time $t_c(s)$ of Theorem 4.1, analytic in both phases and continuous through the EP (circles: numerical onset). Indivisibility is present on both sides; backflow is not.

quantity	scaling	fitted slope
$\mathrm{cond}(\eta)$	$\varepsilon^{-1} \sim \kappa^{-2}$ (Petermann [21])	-0.998
$\mathrm{cond}(\rho)$ (beable-map anisotropy)	$\varepsilon^{-1/2} \sim \kappa^{-1}$	-0.499
$\|\Gamma_\varepsilon(t) - \mathbf{1}\|$ at fixed t	$\varepsilon \sim (\kappa t)^2$	$+0.999$

At fixed laboratory time the metric-corrected process therefore *freezes*: $h \to \cos\theta\,\mathbf{1}$ and $\Gamma_\varepsilon(t) \to \mathbf{1}$. A regularized lab-time limit exists but is trivial—and is not the EP dynamics, which grows like t^2. On the critically rescaled clock $\tau = \kappa t$ the process converges (to 10^{-10} between $\varepsilon = 10^{-4}$ and 10^{-6}) to the universal indivisible Rabi process

$$\Gamma_{\lim}(\tau) = \begin{pmatrix} \cos^2\tau & \sin^2\tau \\ \sin^2\tau & \cos^2\tau \end{pmatrix}, \tag{11}$$

with Rabi axis exactly σ_x; but the beable map identifying its states with the original configurations diverges ($\mathrm{cond}\,\rho \sim \kappa^{-1}$): the survivor is abstract.

By contrast, every open-system quantity of Sec. 4 (reduced-process entries, survival, indivisibility generator floor) varies smoothly and monotonically through s_*. *The EP is a singularity of the similarity-to-Hermitian dictionary, not of the stochastic process*: existence of an indivisible process is governed by the dilation, and the deflation of Proposition 2.1 fails at the EP only because its coordinates do.

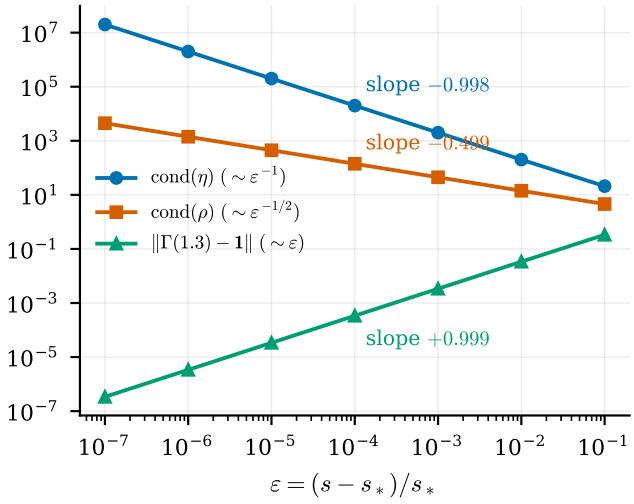


Figure 4: Collapse of the closed-system (metric) description approaching the exceptional point from the unbroken side, $s = s_*(1+\varepsilon)$. The metric condition number diverges with the Petermann scaling $\varepsilon^{-1} \sim \kappa^{-2}$, the beable-map anisotropy as $\varepsilon^{-1/2} \sim \kappa^{-1}$, while the metric-corrected process at fixed laboratory time *freezes*, $\|\Gamma_\varepsilon(t) - \mathbf{1}\| \sim \varepsilon \sim (\kappa t)^2$.

Part II
The ghost problem

6 The Pais–Uhlenbeck oscillator

The PU Lagrangian $L = \frac{\gamma}{2}\left[\ddot{x}^2 - (\omega_1^2+\omega_2^2)\dot{x}^2 + \omega_1^2\omega_2^2 x^2\right]$ has the Ostrogradski Hamiltonian (with $q_1 = x$, $q_2 = \dot{x}$)

$$H_{\mathrm{PU}} = p_1 q_2 + \frac{p_2^2}{2\gamma} + \frac{\gamma}{2}(\omega_1^2+\omega_2^2)q_2^2 - \frac{\gamma}{2}\omega_1^2\omega_2^2 q_1^2, \tag{12}$$

which is *Hermitian* on the real configuration space $(x, \dot{x})$, with the ghost spectrum $E = \omega_1(n_1 + \frac{1}{2}) - \omega_2(n_2 + \frac{1}{2})$: real, unbounded below [9, 8].

6.1 The ghost realization is a valid fixed-beable process

On a discretized $(x, \dot{x})$ grid ($44^2 = 1936$ configurations, spectral derivatives, half-width $L = 10$, $\omega_1 = 1$, $\omega_2 = 0.5$, $\gamma = 1$, and—where a quartic term appears below—$\lambda = 0.05$) the operator (12) is exactly Hermitian; $\Gamma = |U|^2$ is doubly stochastic to 2×10^{-14} and genuinely indivisible (divisor (2) minimum entry -24.55; the family does not compose, $\left\|\Gamma(2t) - \Gamma(t)^2\right\| \approx 2.7$); Hermiticity makes (K1)–(K4) hold trivially with $\eta = \mathbf{1}$, so the beables are fixed. The realization is validated dynamically: Ehrenfest is exact for quadratic Hamiltonians, and $\langle x(t)\rangle$ tracks the exact classical fourth-order trajectory to 3×10^{-3} over half the evolution window (residual from packet tails at the box walls; Fig. 5).

The spectrum of the grid operator runs from -26 to $+140$ and is *completely irrelevant to the process*: at the free level the Ostrogradski ghost is an energy-accounting pathology, invisible to the stochastic representation.

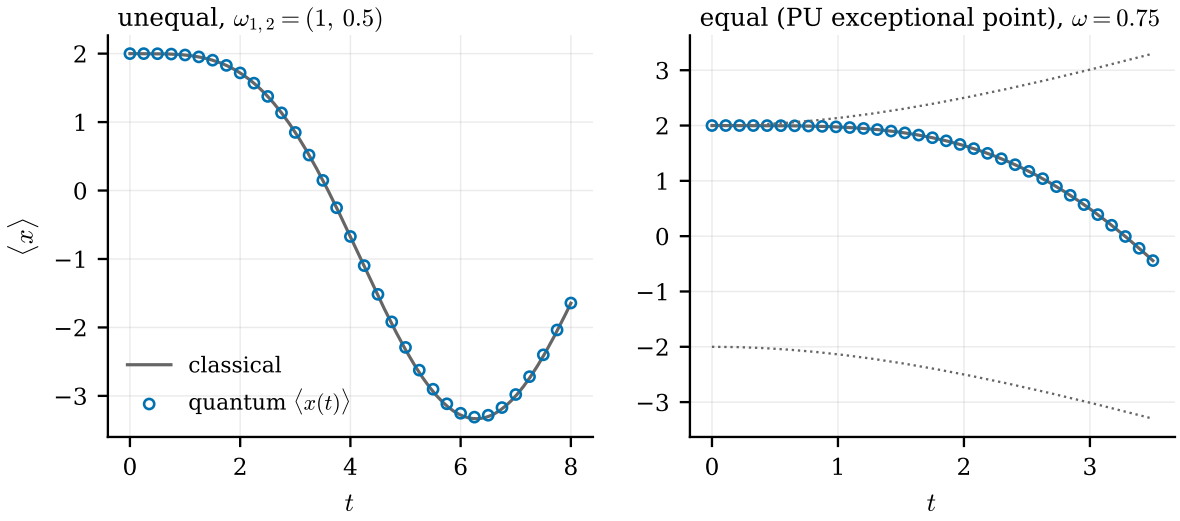


Figure 5: The ghost realization of the free Pais–Uhlenbeck oscillator as a stochastic process on physical configurations: $\langle x(t) \rangle$ of the grid process (circles) versus the exact classical fourth-order trajectory (line). Left: unequal frequencies. Right: equal frequencies—the PU exceptional point—where the classical normal-mode matrix is defective and the solution grows secularly, $x(t) = 2\cos\omega t + \omega t \sin\omega t$ (dotted: envelope $\pm\sqrt{4 + (\omega t)^2}$); the Hermitian realization crosses the point smoothly while Γ remains exactly doubly stochastic.

At equal frequencies ($\omega_1 = \omega_2 = 0.75$; the PU exceptional point) the classical normal-mode matrix is defective (eigenvector condition number $\sim 10^8$; secular $t\sin\omega t$ growth), yet the Hermitian realization passes through smoothly: $\langle x(t) \rangle$ tracks $2\cos\omega t + \omega t \sin\omega t$ to 9×10^{-4} and Γ stays exactly doubly stochastic—the PU-scale confirmation of Sec. 5.

6.2 The no-ghost quantization exits the sample space

Mannheim's positive spectrum requires re-quantizing the ghost mode with a vacuum satisfying $b^\dagger \Omega = 0$, i.e. $\Omega \propto \mathrm{e}^{+y^2/2}$: normalizable only on a rotated contour [8]. This is a *domain change*, not a similarity (a similarity cannot flip the spectrum), and it has no shadow at any truncation: the null vector of the truncated b is $|0\rangle$ at every N (the true vacuum, truncation-stable), while the null vector of the truncated $b^\dagger$ is $|N-1\rangle$, pinned to the truncation edge and escaping to infinite energy; restricted to any fixed low-energy sector, $\sigma_{\min}(b^\dagger) = 1$ exactly. The contour-rotation operator $R = \mathrm{e}^{(\pi/2)D}$, $D = \frac{1}{2}(yp + py) = \frac{\mathrm{i}}{2}(a^{\dagger 2} - a^2)$, has $\log_{10}\mathrm{cond}(R_N)$ reaching ≈ 53 decades by $N = 48$ (tail slope ≈ 1.2 decades per Fock level; the low levels grow slower), so the would-be metric $\sim (RR^\dagger)^{-1}$ is unbounded with unbounded inverse: the Sec. 5 collapse at every scale at once, with no critical-rescaling survivor.

Verdict. The stochastic–quantum correspondence and the $\mathcal{PT}$ no-ghost construction resolve the free ghost in incompatible currencies: Barandes keeps real beables and pays in negative energies; Mannheim buys positive energies and pays with the sample space itself. (A related construction by Salvio and Strumia [43] obtains positive energies and normalizable wave functions at the price of a negative-norm configuration-space inner product—again incompatible with a probability measure over configurations.) The dynamics-dependent beable rotation of

Sec. 2 escalates, in the PU limit, to beable *non-existence*.

7 Interacting ghosts: conservative, transient, explosive

Does the ghost instability, once interactions are switched on, become a *stochastic* pathology? The answer arrives in stages.

7.1 Quadratic interactions: a valid process turns transient

Consider the Hermitian cascade Hamiltonian

$$H = \omega_1 a^\dagger a - \omega_2 b^\dagger b + g\,(a^\dagger b^\dagger + ab), \tag{13}$$

whose normal twin has $+\omega_2 b^\dagger b$. The Heisenberg mode matrix has eigenvalues $\bar\omega \pm \sqrt{\delta^2/4 - g^2}$ (ghost; $\delta = \omega_1 - \omega_2$, $\bar\omega = (\omega_1 + \omega_2)/2$), so the ghost sign collapses the instability threshold from the *sum* frequency, $g > (\omega_1 + \omega_2)/2$ (normal), to the *difference* frequency, $g > |\omega_1 - \omega_2|/2$: zero at resonance. At resonance the vacuum cascades with the exact two-mode-squeezing laws

$$\langle n_1(t)\rangle = \langle n_2(t)\rangle = \sinh^2(gt), \qquad P\big(n_1 \le K\big) = 1 - \tanh^{2K+2}(gt) \;\longrightarrow\; 0, \tag{14}$$

both matched numerically to $\lesssim 10^{-3}$ (four decimals for $t \le 6$) in the resolved regime ($\langle n\rangle \lesssim$ a few, before the 34^2 Fock truncation sets in at the last tabulated time) (Fig. 6; Fock truncation 34^2, $g = 0.15$; detuned growth consistent with $(g^2/\kappa_g^2)\sinh^2\kappa_g t$, $\kappa_g = \sqrt{g^2 - \delta^2/4}$), while Γ remains doubly stochastic to 2×10^{-14} and the energy books stay exactly balanced ($E_{\rm normal} = +8.301$, $E_{\rm ghost} = -8.301$ at $t = 12$, where the 34^2 truncation already costs $\sim$4% against the exact $\sinh^2(gt)$; $\langle H\rangle = 0$ is conserved to 4×10^{-15} throughout, and it is that balance, not the individual energies, that the argument uses).

Vacuum decay is a configuration-space cascade with balanced energy books: the process is valid and indivisible but *transient*—it permanently escapes every finite region at rate $2g$—which is the first stage at which the ghost becomes visible in the process's own currency.

7.2 Malicious nonlinearities: finite-time escape

Classically, the ghost pair $\frac{1}{2}(p_1^2 + \omega_1^2 x_1^2) - \frac{1}{2}(p_2^2 + \omega_2^2 x_2^2) + \lambda x_1^2 x_2^2$ is mostly bounded at moderate amplitude (benign in the sense of [11]), while the quartic PU,

$$\dddot{x} + (\omega_1^2 + \omega_2^2)\ddot{x} + \omega_1^2\omega_2^2 x = \frac{4\lambda}{\gamma}x^3, \tag{15}$$

exhibits robust *finite-time* blowup for both signs of λ at every amplitude tested ($t_* = 3.3$–6.9); for $\lambda > 0$ the dominant balance is $x \sim A(t_* - t)^{-2}$, $A = \sqrt{30\gamma/\lambda}$ (the $\lambda < 0$ escape proceeds by a different mechanism, taken up in Sec. 10). Quantum-mechanically, a lattice can transport the accelerating wavefront only while the required classical speed stays below the Nyquist momentum $\pi/\Delta x$; box-size drifts of the leak onset correlate exactly with $v_{\rm cl}({\rm edge})/k_{\max} = 1.4, 2.4, 3.4$ and are lattice confinement, not physics (a fact that itself echoes Sec. 8: a non-essentially self-adjoint operator has no faithful finite truncation). In the resolved regime, quantum probability arrives at every probed radius ($r \le 7$; the $r = 8$ probe is marginally super-Nyquist) at times bounded by the classical finite-time law, with bulk weight, while the free control never passes $r \approx 7$.

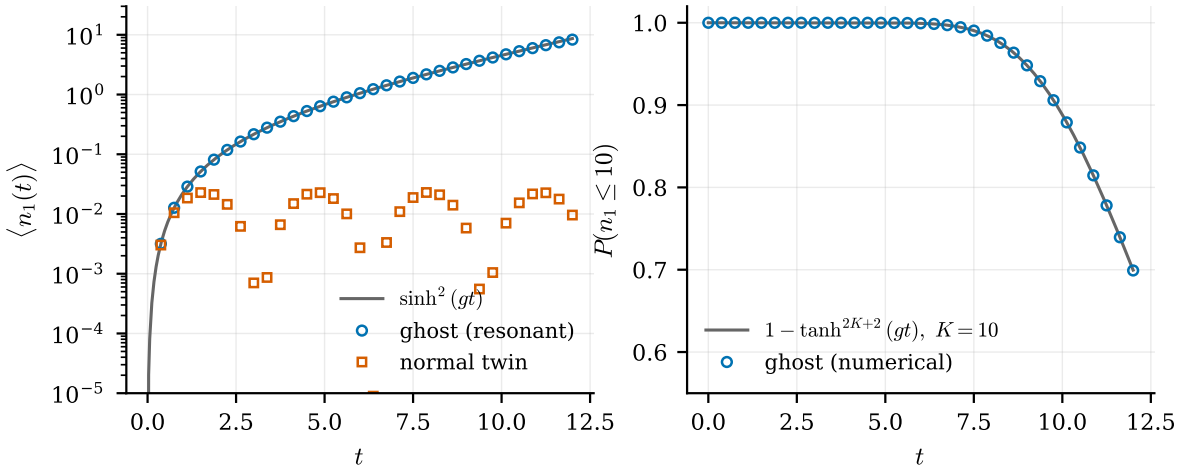


Figure 6: The quadratic ghost cascade (13) at resonance ($\omega_1 = \omega_2$, $g = 0.15$): a valid, exactly doubly stochastic process that is *transient*. Left: occupation growth $\langle n_1(t)\rangle = \sinh^2(gt)$ (line: exact law; circles: Fock-space numerics), while the normal-sign twin (squares) stays bounded below 3×10^{-2}—the instability threshold collapses from the sum to the difference frequency. Right: escape from the occupation ball $n_1 \leq 10$ follows the exact two-mode-squeezing law (14).

In the standard vocabulary of stochastic processes, the hierarchy is: free/normal-sign = *conservative*; ghost with quadratic (or benign nonlinear) interaction = *transient*; ghost with malicious nonlinearity = *explosive* (mass leaves the state space in finite time)—the last verdict at the level of the classical mechanism (its minimal escape process, Sec. 8); the quantum operator can resist (the $\lambda < 0$ completion candidates of Sec. 10). The Ostrogradski ghost is a recurrence-classification problem. Complementary operator-theoretic and dynamical stability results for interacting ghosts—the global-stability classification program of Errasti Díez, Gaset Rifà, and Staudt [44] and its multidimensional classical-scattering extension by Held [45], conserved quantities with discrete spectrum [46], and exact moment bounds [47]—identify interaction structures in which the cascade is arrested; in the present language, those are mechanisms enforcing recurrence (none of these works uses the stochastic/recurrence currency itself).

8 The explosion dictionary

The mechanism of malicious escape is one-dimensional: motion along a direction where the effective potential falls super-quadratically. For this mechanism the correspondence between the three descriptions can be made a theorem.

Theorem 8.1 (explosion dictionary, 1D mechanism model)**.** *Let $H = \frac{1}{2}p^2 + V$ on the line with an escaping end ($V \to -\infty$ eventually monotonically and C^1 with $|V'|\,|V|^{-3/2}$ bounded (the mechanism model), the standard regularity under which the limit-*point *criterion below applies; the limit-circle leg (2) additionally requires the C^2 error control ($|V''|\,|V|^{-3/2}$ integrable) stated there. Rauch–Reed's one-dimensional counterexamples [36] show these hypotheses cannot be dropped), and define the classical time of flight $T(E) = \int^{\infty} \mathrm{d}x/\sqrt{2(E - V(x))}$.*

(1) If $T = \infty$ (e.g. $V \gtrsim -Kx^2$), the end is limit point (Levinson's monotone criterion [24]; the

power-law case also by [23]), and—with the other end also limit point (the non-escaping end confining, or a second escaping end likewise at $T = \infty$, as in $V = -x^2$)—H is essentially self-adjoint [23], there is a unique unitary $U(t)$ and a unique conservative process.

(2) *If $T < \infty$ (e.g. $V \lesssim -Kx^{2+\epsilon}$) and $V \in C^2$ with $|V''|\,|V|^{-3/2}$ integrable (the added control of Prop. C.4, automatic for the power-law potentials here), the end is limit circle [23], by the (WKB-asymptotic) identity*

$$|\psi_\pm(x)|^2 \;\approx\; \big[2(E - V(x))\big]^{-1/2} \quad\Longrightarrow\quad \int^\infty |\psi_\pm|^2 \,\mathrm{d}x \;\approx\; T\,: \tag{16}$$

the deficiency solution's norm is, to WKB accuracy, the classical time of flight.[1] *This added control is what makes both the limit-circle classification and its quantitative norm identity rigorous—under C^1 boundedness alone we expect a monotone staircase potential can have $T < \infty$ yet be limit point, so the smoothness should not be dropped. Consequently:*

- *(a) there is no canonical unitary dynamics: self-adjoint extensions form a $U(n)$ family for n escaping ends ($U(1)$ for one, $U(2)$ for two, including completions that transmit through infinity), i.e. boundary data at infinity [23];*
- *(b) every extension is conservative but, semiclassically, returns probability from infinity after the finite classical round trip; its spectrum is discrete with local spacing, semiclassically, $\Delta E \simeq \pi/T_{\mathrm{cross}}(E)$, T_{cross} the classical traversal time;*
- *(c) the regulator-independent object is the minimal (Feller) semigroup [27] of the* escape diffusion $\mathrm{d}X = \sqrt{2(E - V(X))} \circ (\mathrm{d}t + \varepsilon\,\mathrm{d}W)$*—the classical flow run with a Brownian-perturbed clock, whose speed measure (driftless variant) is twice the WKB deficiency density of (16) (Lemma 8.2)—a sub-stochastic,* explosive *process losing mass at the classical escape time;*
- *(d) one integral decides classical completeness, quantum essential self-adjointness (Weyl [22]), and stochastic conservativity (Feller boundary classification: $+\infty$ is regular—accessible—when $T < \infty$ and natural when $T = \infty$; Lemma 8.2).*

The rigorous inputs are Weyl's alternative and the limit-point/limit-circle criteria of [24, 23] (Thms. X.7–X.10); the WKB norms and the semiclassical spacing law are used at their standard validity.

The stochastic leg (2)(c)–(d) is not carried by H itself: in the malicious case every self-adjoint extension has spectrum unbounded below, so e^{-tH} is not even bounded, let alone sub-Markov, and $V \to -\infty$ is neither a drift nor a nonnegative killing rate; nor is a ground-state (h-)transform available, since $E - V \geq c > 0$ on the escaping end makes $u'' = -2(E - V)u$ oscillatory there, with no eventually-positive solution. The leg is carried instead by the diffusion whose scale and speed are read off the classical flow.

Lemma 8.2 (the stochastic leg: an escape diffusion)**.** *Under the hypotheses of Theorem 8.1 (in fact only V continuous with $V \to -\infty$ is used), fix E and x_0 with $\sup_{x \geq x_0} V < E$, and set $v(x) = \sqrt{2(E - V(x))}$, $\tau(x) = \int_{x_0}^x \mathrm{d}y/v(y)$, $T = T(E) = \lim_{x\to\infty} \tau(x)$. Let X be the escape*

[1] Here $\psi_\pm$ (and, in Theorem B.2(iii), u_n) denote the L^2 solutions at the *real* mechanism energy E; the genuine deficiency vectors solve $(\tau \mp \mathrm{i})u = 0$, and their WKB modulus carries an extra factor $\exp\big(\mp 2\,\mathrm{Im}\int^x \sqrt{2(z - V)}\big) = e^{O(T)}$, bounded whenever $T < \infty$. The limit-circle *classification* (finiteness of the integral) is therefore unaffected, while the quantitative norm identity "$\approx T$" holds up to this bounded (x-independent, though T-dependent) factor.

	$V = -x^4$ (malicious)		$V = -x^2$ (marginal)	
L	measured ΔE	π/T_{cross}	measured ΔE	π/T_{cross}
6	2.559	2.609	1.619	1.640
9	2.365	2.449	1.290	1.289
12	2.345	2.377	1.106	1.112
16	2.318	2.325	0.964	0.975

Table 2: Level spacing of the wall-regularized completions near $E = 11$ versus the classical law $\pi/T_{\text{cross}}(E, L)$. For the malicious potential the traversal time to infinity is finite, so the spacing saturates: every conservative completion reflects off infinity at a finite recurrence time. For the marginal control the spacing decays toward the continuum.

diffusion $\mathrm{d}X = v(X) \circ (\mathrm{d}t + \varepsilon\, \mathrm{d}W)$ *(Stratonovich, any* $\varepsilon > 0$*) or its driftless variant* $\mathrm{d}X = v(X) \circ \mathrm{d}W$*, reflected at* x_0 *and minimal at* $+\infty$*, with explosion time* $\zeta = \inf\{t : X_{t-} = +\infty\}$ *and minimal semigroup* $(P_t^{\min})$*; in scale/speed form the driftless variant has* $s(x) = \tau(x)$ *and* $m(\mathrm{d}x) = 2\,\mathrm{d}x/v(x)$*, i.e.* m *is twice the WKB deficiency density of* (16)*, and its scale mass at* $+\infty$ *is* T*. Then:*

(i) $+\infty$ *is a* regular *boundary if* $T < \infty$ *and a* natural *boundary if* $T = \infty$*; it is never exit or entrance. For the driftless variant the two Feller test integrals coincide,* $\Sigma = N = T^2$*.*

(ii) If $T < \infty$ *then* $\mathbb{P}_x(\zeta < \infty) = 1$*:* $(P_t^{\min})$ *is strictly sub-stochastic (*$P_t^{\min}\mathbf{1} < 1$ *for* $t > 0$*), the mean explosion time is* $\mathbb{E}_{x_0}[\zeta] = \Sigma$ *(*$= T^2$ *driftless), and* $\zeta \to T$ *almost surely as* $\varepsilon \downarrow 0$ *(drifted variant). If* $T = \infty$ *then* $(P_t^{\min})$ *is conservative and,* $+\infty$ *being natural, unique.*

(iii) Consequently one integral decides all three legs; the diffusion's own Weyl classification agrees; and at the regular boundary Feller's lateral conditions (absorbing, reflecting, elastic, sticky, jump-in) parametrize the sub-Markov extensions of $(P_t^{\min})$*—the stochastic mirror of the* $U(1)$ *family of self-adjoint extensions in (2)(a), the conservative ones returning probability from infinity as in (2)(b).*

The proof constructs the escape diffusion explicitly and reduces the Feller boundary classification to the single classical integral $T(E)$ (in the clock coordinate both variants are Brownian motion on $[0, T)$, and $\Sigma = N = T^2$ for the driftless one); it is given in Appendix A.

Numerical verification (representative $V = -x^4$—the two-escaping-end case, deficiency $(2, 2)$, the $U(2)$ completions of (2)(a)—and marginal control $V = -x^2$, limit-point with deficiency $(0, 0)$): the spacing law matches to 3.5% or better in all eight (V, L) cases tested and to 0.3% at the largest box for $V = -x^4$ (Table 2); the eigenvalue nearest $E = 10$ (a tracked-level probe, distinct from the spacing window at $E = 11$) sweeps erratically with the wall position (no completion has an $L \to \infty$ limit) while the spacing is stable; a 1D absorbing-boundary evolution (where the lattice barrier is beatable) absorbs 98.6–98.8% of the probability by $t = 1$ *independently of box size*, with onset saturating toward the classical arrival ($T_{\text{esc}} = 0.4635$, released from rest at $x = 2$, i.e. $E = -16$; Fig. 7, left); and two finite-box stand-ins for the extreme completions $R = 1, 0$—Dirichlet wall versus periodic identification—both revive $\approx 95\%$ of the probability after the finite round trip but from *opposite sides* (reflection versus a $+\infty \to -\infty$ "wormhole"; Fig. 7, right), an observable difference encoded nowhere in H.

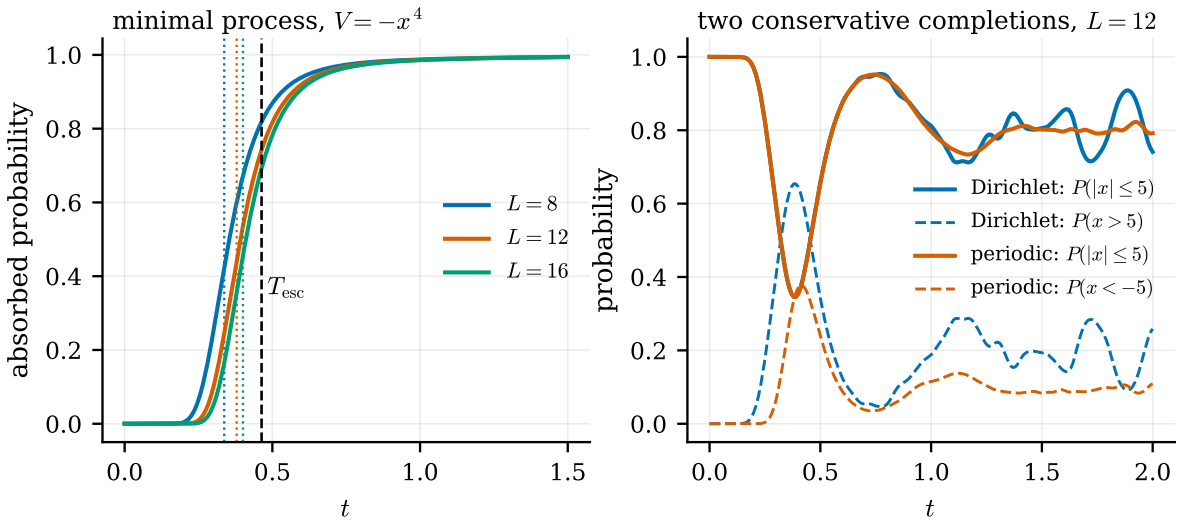


Figure 7: Explosion and its completions for $V = -x^4$ (the explosion dictionary, Theorem 8.1). Left: the minimal (absorbed) process is box-independent—the absorbed probability for $L = 8, 12, 16$ collapses onto one curve with onset tracking the classical arrival at each absorber edge (dotted verticals) and saturating at the finite escape time T_{esc} (dashed): probability reaches infinity at finite time. Right: two finite-box (wall-regularized) stand-ins for the extreme completions $R = 1, 0$. Both revive $\approx 95\%$ of the probability after the finite round trip "through infinity," but the Dirichlet wall returns it from the right (reflection) while the periodic identification returns it from the left (transmission, $+\infty \to -\infty$): the completion is boundary data at infinity, not physics contained in H.

9 The completions dictionary: self-adjoint extensions and Feller lateral conditions

Theorem 8.1(2)(a)–(b) left the conservative physics of a malicious end distributed over a family of completions—a $U(n)$ of boundary conditions at infinity for n escaping ends ($U(1)$ for one, a $U(2)$ for two), including the completions that transmit through infinity (the "wormhole" of Fig. 7, right)—while Lemma 8.2(iii) recorded the parallel fact on the stochastic side: at the regular boundary, Feller's lateral conditions parametrize the sub-Markov extensions of the minimal semigroup. We called the second family the stochastic mirror of the first. This section makes the mirror a theorem and, in doing so, determines exactly what kind of mirror it is. It is *not* a bijection, and should not be read as one: the correspondence is a surjection of the unitary family onto a small, exactly characterized slice of the Feller–Wentzell cone, with the quantum phases collapsing fiberwise. The two-sided failure of bijectivity is itself the physics—the quantum side carries phase (interference/spectral) data invisible to probability, and the stochastic side carries dissipative and boundary-mass data forbidden to unitarity. Everything at the class level below is exact, by limit-circle boundary-form algebra whose only WKB input is the Olver-normalized in/out frame that pins the boundary basis (App. C); only the wave-packet arrival/revival interpretation is semiclassical, and it is flagged as such there.

Fix the mechanism energy E—the same E at which the escape diffusion of Lemma 8.2 is built—and let $n \in \{1, 2\}$ count the escaping ends with finite time of flight ($n = 1$: the other

end confining, hence limit point with no boundary data, deficiency $(1,1)$; $n=2$: e.g. $V=-x^4$, the deficiency-$(2,2)$ case of the numerics). At each escaping end the solutions of $\tau u = Eu$ carry a conserved probability current of signature $(1,1)$, splitting each germ into a current-normalized *incident* amplitude a_i and a *returned* amplitude b_i. The first result (Proposition C.1, Appendix C) is that for every self-adjoint extension $\tilde{H}$ the admissible germs at energy E satisfy $b = S_{\tilde{H}}(E)\,a$ for a unique unitary $S_{\tilde{H}}(E) \in U(n)$, and $\tilde{H} \mapsto S_{\tilde{H}}(E)$ is a *bijection* from the self-adjoint extensions onto $U(n)$: the completion literally is a boundary S-matrix, the n-channel instance of the S-matrix at infinity of Sec. 10. Probability, however, can see only the flux shadow of $S(E)$: the *flux functor* Φ_E (Definition C.3) assigns to $\tilde{H}$ the lateral datum "reflect at end i with weight $|S_{ii}(E)|^2$, jump to the other end with weight $|S_{ij}(E)|^2$; no killing, no stickiness."

Theorem 9.1 (extension–lateral dictionary; $n \in \{1,2\}$ escaping ends, $T_i(E) < \infty$, Olver-frame hypothesis of App. C; class level proved)**.** *Let $\mathcal{C}$ be the set of Feller–Wentzell lateral data (Eq. (19); one datum per escaping end) that are simultaneously* (c1) conservative *(no killing),* (c2) instantaneous *(no boundary stickiness),* (c3) boundary-supported *(the jump measure charges only the other escaping end, never the interior), and* (c4) reciprocal *(equal passage weights in the two directions). Then:*

(1) ***(image)*** *Φ_E maps the self-adjoint extensions of H* onto *$\mathcal{C}$ and* only into *$\mathcal{C}$; the correspondence is not a bijection in either direction.*

(2) ***($n=1$: total collapse)*** *$\mathcal{C}$ is a single point—pure reflection, $\mathrm{d}u/\mathrm{d}s\,(+\infty) = 0$—and Φ_E is the* constant *map: the entire $U(1)$ of completions has one stochastic shadow. The phase fixes the eigenvalue positions (interference data) and has no sub-Markov counterpart.*

(3) ***($n=2$: a segment)*** *$\mathcal{C} \cong [0,1] \ni R = |S_{11}(E)|^2$: reflect with weight R, pass through infinity ($+\infty \to -\infty$ and $-\infty \to +\infty$) with weight $1-R$, necessarily the* same *R at both ends. $R=1$ are the separated completions (e.g. the Dirichlet wall); $R=0$ are the wormhole/gluing completions of Theorem 8.1(2)(a), whose stochastic shadow is Feller's circle-type passage: the jump-in datum linking $+\infty$ to $-\infty$ (the "darning" of the two boundary points). The fiber over each R is a phase torus generically of dimension 3 ($\dim U(2) = 4$ against $\dim \mathcal{C} = 1$).*

(4) ***(sub-unitary $\leftrightarrow$ sub-Markov)*** *The flux functor lands in $\mathcal{C}$ ($\operatorname{im}\Phi_E \subseteq \mathcal{C}$); conversely, no lateral datum* outside *$\mathcal{C}$ admits a correspondence to a self-adjoint extension that preserves total probability, the state-space measure class, and the boundary flux data, each obstruction enforced by an exact invariant (Appendix C). In particular, killing (absorbing, elastic) is matched not by self-adjoint but by* maximal-dissipative *completions: the contractive boundary data form the closed matrix ball $\|S(E)\| \le 1$, of which the self-adjoint family is its unitary part—the extreme points (for $n=1$, the topological boundary); $S=0$ (the outgoing/radiation condition $b=0$) is the absorbing datum—precisely the minimal explosive semigroup of Theorem 8.1(2)(c), the completion* missing *from the self-adjoint family—and $0 < \|S\| < 1$ are the elastic data. Sticky and interior-jump data have no counterpart in $L^2(\mathrm{d}x)$ at all.*

The proof, entirely boundary-form algebra, is given in Appendix C, together with the full parametrization and the four exclusions each tied to its invariant.

The $n=1$ collapse is structural, not an artifact of the map chosen: on the escaping end $v \ge v_{\min} > 0$, so the quantum phase $\int^\infty v\,\mathrm{d}x$ diverges while the clock $\int^\infty \mathrm{d}x/v = T$ converges.

The quantum boundary basis winds infinitely often approaching the end, whereas the diffusion's harmonic boundary basis $(1, s)$ is monotone with finite limits; no linear identification of the two boundary spaces can be canonical, and the only basis-independent quantities transferable to the probabilistic side are the current moduli $|S_{ij}|^2$. The phase is precisely the datum the $|\cdot|^2$ functor forgets (Remark C.6).

One caveat deserves the main text rather than a footnote: the dictionary is a correspondence between two *parallel parametrized families*—completions of H in $L^2(I, \mathrm{d}x)$ on one side, lateral data of the escape diffusion on the other—a structural mirror, not an isomorphism of spaces or of dynamics. No Hilbert-space map between the two evolutions is asserted or possible; what transfers is exactly the flux/class data, and Theorem 9.1 says this transfer is onto the slice $\mathcal{C}$ and nothing more. Likewise the dictionary is stated at the mechanism energy E, the same E at which the diffusion itself is defined (a matching of reference energies intrinsic to the construction); for a fixed completion the weight R is generically E-dependent, the separated and smooth-gluing families being the E-uniform exceptions (Remark C.2), and at $\varepsilon > 0$ the reflect/jump weights are an excursion-rate convention, literal per-arrival probabilities only in the transport limit $\varepsilon \downarrow 0$ (Remark C.5).

The revival numerics of Fig. 7 (right) are the two extreme points of the segment: the Dirichlet wall is $R = 1$ (return from the right), the periodic identification is $R = 0$ (return from the left, through the wormhole), and intermediate completions split the returning probability with weights R and $1-R$ at the same finite round trip. That wave-packet statement is the semiclassical leg of the dictionary—the only leg that is (Proposition C.4, *[proved, sketch]*).

10 Multidimensional theory: channels and the S-matrix at infinity

The naive dictionary "classical incompleteness $\Leftrightarrow$ quantum non-self-adjointness" fails in general beyond the monotone one-dimensional mechanism model: Rauch and Reed constructed one-dimensional counterexamples in both directions [36], and in $n \geq 2$ the correct structure is channel-wise. Throughout, $h_\perp$ has C_0^∞ in its domain (as for any Schrödinger-type transverse operator), H is defined on C_0^∞, and deficiency indices refer to its closure.

Theorem 10.1 (quadratic ghosts are safe)**.** *Every at-most-quadratic Hamiltonian on $\mathbb{R}^n$—including the indefinite cascade (13) at any coupling, even in the unstable phase, and the free PU operator (12)—is essentially self-adjoint on the span of Hermite functions.*

Proof. A quadratic H maps Hermite level N into levels $\leq N + 2$ with coefficients $O(N)$, so $\|H^n \psi_N\| \leq (2C)^n \, (N/2+n)!/(N/2)!$ and $\sum_n t^n \, \|H^n \psi_N\| \, /n!$ converges for small t: every Hermite function is an analytic vector. Nelson's analytic vector theorem [26] applies. □

Thus instability never breaks uniqueness or conservativity: the transient/explosive dichotomy of Sec. 7 is rigorous on the transient side.

Theorem 10.2 (channel additivity of deficiency)**.** *Let $H = h_1 \otimes \mathbf{1} + \mathbf{1} \otimes h_\perp$ on $L^2(\mathbb{R} \times \mathbb{R}^m)$, with $h_1 = \frac{1}{2}p^2 + V_1$ limit circle at both ends $\pm\infty$ (e.g. V_1 super-quadratically falling with the C^2 error-control of Theorem 8.1) and $h_\perp$ self-adjoint with orthonormal eigenbasis $\{\varphi_n, E_n\}$. Then for every n and every solution u of $\big(h_1 - (\pm\mathrm{i} - E_n)\big)u = 0$ (all of which are L^2, since the limit-circle property at each end is independent of the spectral parameter), $u \otimes \varphi_n \in \ker(H^* \mp \mathrm{i})$. Hence the deficiency indices of H are (∞, ∞).*

Proof. H^* acts distributionally, and $H(u\otimes\varphi_n) = \big((\pm\mathrm{i}-E_n)u\big)\otimes\varphi_n + E_n\,u\otimes\varphi_n = \pm\mathrm{i}\,u\otimes\varphi_n \in L^2$, so $u\otimes\varphi_n \in D(H^*)$ and is a deficiency vector; vectors with different n are orthogonal. □

Corollary 10.3. $-\frac{1}{2}\Delta - c|x|^\alpha$ *on* $\mathbb{R}^d$, $d \geq 2$, *with* $c > 0$, $\alpha > 2$ *has deficiency indices* (∞, ∞) *(the same channel argument as Theorem 10.2, now applied to the angular-momentum decomposition rather than a literal tensor sum: each fixed-*ℓ *radial channel is limit circle at infinity—the super-quadratic* $-c|x|^\alpha$ *fall governs each end (*C^2 *automatic for the power law), the subquadratic centrifugal term not spoiling it—and there are infinitely many channels).*

Theorem 10.4 (non-separable channel mixing)**.** *Let* $H_0 = \frac{1}{2}p_2^2 - |x_2|^\alpha + h_\perp(x_1)$, $\alpha > 2$, *with* $h_\perp$ *having orthonormal eigenbasis as in Theorem 10.2 (so that* H_0*'s indices are infinite), and* $B = \mu f(x_1)g(x_2)$ *with* f, g *bounded and real-valued. Then* $H_0 + B$ *has the same (infinite) deficiency indices as* H_0*, while* B *genuinely mixes the transverse channels.*

Proof. Bounded symmetric perturbations preserve deficiency indices [25]. □

Physical content. The conservative completions of a malicious multidimensional ghost are parametrized by unitaries between infinite-dimensional deficiency spaces: a genuine *S-matrix at infinity*, which may scramble the transverse state upon "reflection from infinity". Numerically (four-channel model, $V_1 = -|x|^{5/2}$, bounded mixing): the wall-regularized level count in a window matches the channel sum $\sum_n \int T_{\mathrm{cross}}(E - E_n, L)\,\mathrm{d}E/\pi$ (34 vs 32.3 at $L = 8$; 37 vs 37.1 at $L = 12$), at $\mu = 0$ the spectrum is exactly a union of shifted 1D ladders (deviation 10^{-12}), switching on the bounded coupling leaves the counts essentially unchanged ($34 \to 34$ at $L = 8$, $37 \to 36$ at $L = 12$, a one-level window-edge shift within count granularity; Kato stability made visible)—and a packet sent to infinity in the transverse ground state returns as a channel mixture *whose content depends on where infinity was regularized* (Fig. 8): fractions $(0.583, 0.346, 0.058, 0.014)$ for $L = 6$ versus $(0.401, 0.421, 0.143, 0.035)$ for $L = 9$, with 96%/93% of the probability returning at the classical round-trip times.

Remark 10.5 (an obstruction specific to ghosts). For indefinite $H_0 = h_1 \otimes \mathbf{1} + \mathbf{1} \otimes h_\perp$ with h_1 unbounded below, transverse observables are *not* H_0-bounded: joint spectral points $(E_1, E_\perp)$ with $E_1 \approx -E_\perp$ make $\|H_0\psi\|$ small while $\|h_\perp\psi\|$ is large. The standard Kato–Rellich transfer therefore fails even for mildly unbounded couplings—a second, ghost-specific obstruction beyond non-separability.

Open Problem 10.6. Determine the deficiency indices of the exact quartic Pais–Uhlenbeck operator, $H_{\mathrm{PU}} + \lambda q_1^4$ of (12), and of the ghost pair with coupling $\lambda x_1^2 x_2^2$, on $C_0^\infty(\mathbb{R}^2)$. Both escape the hypotheses above (unbounded diagonal coupling; degenerate/indefinite kinetic form). Either outcome is significant: infinite deficiency would complete the explosion dictionary for the physical operators, while essential self-adjointness would be a "quantum completion" in the sense of [36]—interactions curing the ghost's explosion.

Numerical evidence. A regulator-dependence campaign (Fock-space compressions, which regulate position and momentum symmetrically—the PU escape accelerates without bound—with the matched-eigenvalue-drift diagnostic calibrated on limit-circle, continuous-spectrum, and discrete essentially-self-adjoint controls, and on the free PU operator as the two-dimensional baseline) points to a split verdict. The quartic PU shows persistent $O(1)$ drift that does not decay with truncation, at two different basis scales—the limit-circle signature, at $\sim 5\times$ the free-PU baseline: evidence *against* essential self-adjointness, i.e. for the explosion dictionary applying to the physical operator. The ghost pair with $\lambda > 0$ sits at the essentially-self-adjoint baseline,

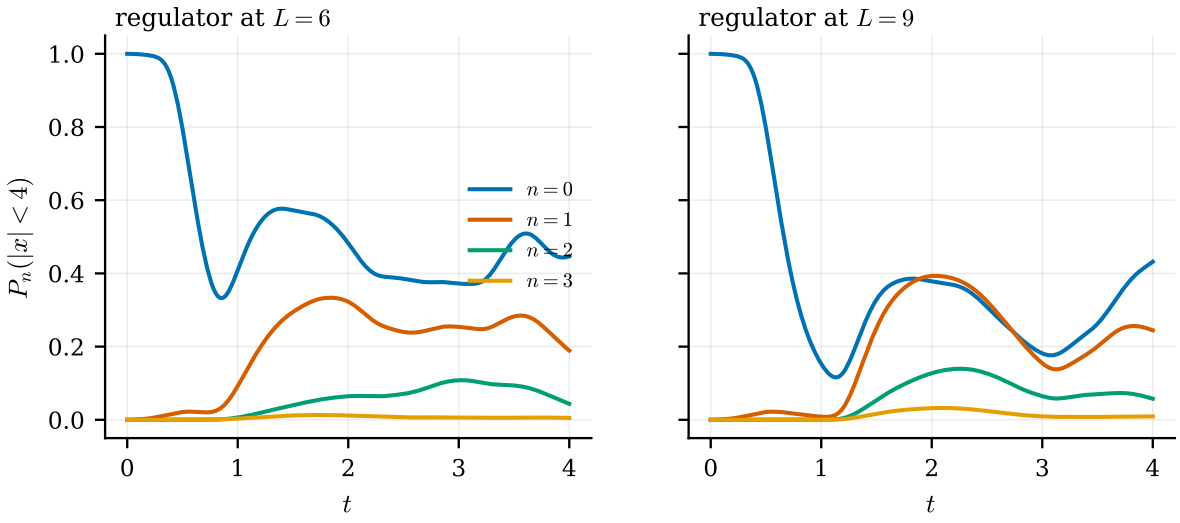


Figure 8: The S-matrix at infinity made visible (four-channel malicious model of the bounded-mixing theorem, Theorem 10.4). Channel-resolved probability in the interior region $|x| < 4$ for a packet launched toward infinity in the transverse ground state $n = 0$. The packet reflects off the regulator and returns as a channel *mixture*; comparing the two regulator positions ($L = 6$, left; $L = 9$, right), the returned mixture differs substantially. For the unregulated operator, conservative completions are genuine unitaries between the infinite-dimensional deficiency spaces (indices (∞, ∞) by Theorem 10.2, parametrization by von Neumann's extension theory); there is no $L \to \infty$ limit of the regulated reflection law—the returned mixture is completion data injected at infinity, which the regulator merely imitates, differently at each L.

as predicted by its Born–Oppenheimer channels $W_n(x_2) = \frac{1}{2}\omega_2^2 x_2^2 - (n + \frac{1}{2})\sqrt{\omega_1^2 + 2\lambda x_2^2}$, all of which are confining. Unexpectedly, so does the classically malicious $\lambda < 0$ ghost pair—tentative evidence of a genuine Rauch–Reed-type quantum completion, in which zero-point energy walls off the narrow classical escape valley. Evidence, not proof; but it sharpens the problem: prove deficiency for the quartic PU by quasi-mode construction along its escape channel, and test the quantum-completion hypothesis for the ghost pair with valley-adapted bases. (Deficiency indices of related polynomial-oscillator operators—higher-order squeezing—have recently been computed rigorously, including cases where added terms *restore* essential self-adjointness [49]: an operator-theoretic precedent for completion by interaction.)

The drift verdict for the quartic PU can, in fact, be given structural support, together with a falsifiable sign prediction:

Proposition 10.7 (displaced-oscillator identity (exact) and channelization (to $O(k^2)$) of the quartic PU)**.** *In the Fourier-rotated representation ($p_1 \to y$, $q_1 \to -p_y$), the displaced-oscillator unitary* $U = \exp[-\mathrm{i}\, y\, p_2/(\gamma\Omega)]$, $\Omega = \omega_1^2 + \omega_2^2$, *gives* exactly

$$U\,(H_{\mathrm{PU}} + \lambda q_1^4)\,U^\dagger = Q\Big(p_y + \frac{p_2}{\gamma\Omega}\Big) + \frac{p_2^2}{2\gamma} + \frac{\gamma\Omega}{2}q_2^2 - \frac{y^2}{2\gamma\Omega}, \qquad Q(p) = \lambda p^4 - \frac{\gamma\Omega_0^2}{2}p^2, \tag{17}$$

with $\Omega_0^2 = (\omega_1\omega_2)^2$ *(the conjugating generator is bilinear, so the Baker–Campbell–Hausdorff series terminates and the identity is exact; confirmed numerically in Fock space to* 10^{-9}*). The*

mode-2-diagonal (channel) part is, in the $k = p_y$ representation and up to $O(k^2)$ corrections, minus *the Schrödinger operator $\hat{p}^2/(2\gamma\Omega) - \lambda k^4 + (\gamma\Omega_0^2/2)k^2$; interchannel couplings are polynomial of degree ≤ 3 in k, subordinate to the quartic diagonal. Hence for $\lambda > 0$ every channel is limit circle (finite channel time of flight; the truncated window $k = 2 \to 400$ at $E_0 = 1$ gives 1.74 for the parameters used here, $\lambda = 0.05$, against 2.06 from the turning point to infinity), and channel additivity as in Theorem 10.2 argues for deficiency indices (∞, ∞)—with the stability question settled at every finite channel number by the finite-channel stability theorem (Theorem 10.8) below. For $\lambda < 0$ every channel is (minus) a confining operator: quantum completion predicted,* despite *classical finite-time blowup at both signs.*

The sign prediction is confirmed by the drift diagnostic: the classically explosive $\lambda < 0$ quartic PU sits near the essentially-self-adjoint baseline (ratios 0.17–0.46 at two basis scales, measured over $N = 24$–42, none leaving the essentially-self-adjoint baseline band), separated from the $\lambda > 0$ limit-circle signature (the extended truncations give $O(1)$ ratios—3.77 at $N = 48 \to 56$, 1.06 at $N = 56 \to 64$; at the matched $N = 24$–42 range the $\lambda > 0$ ratios are 0.79–1.96, so the separation is not an artifact of comparing disjoint windows—several times to an order of magnitude above the ∼0.2 essentially self-adjoint baseline; a single early window at one basis scale dips to 0.34), while the $\lambda < 0$ ghost pair gives 0.49 at $N = 48 \to 56$ and 0.32 at $N = 56 \to 64$—the first value at the upper edge of the essentially-self-adjoint band (above the tighter 2D free-PU control) but both well below the limit-circle signature. The $\lambda < 0$ quartic PU thus joins the $\lambda < 0$ ghost pair as a Rauch–Reed-type quantum-completion candidate: in both cases the classical escape appears to be walled off quantum mechanically, whereas the $\lambda > 0$ quartic PU escapes through momentum space along limit-circle channels.

The stability of channel deficiency under the interchannel couplings—the gap flagged after Proposition 10.7—can in fact be closed at every finite channel number, because the couplings are not generic subordinate couplings: they are functions of the single operator p_2.

Theorem 10.8 (finite-channel deficiency stability)**.** *For every M, the M-channel Galerkin model of the channelized quartic PU ($\lambda > 0$),*

$$S_M = \frac{\hat{p}^2}{2\gamma\Omega} - Q\big(k\,\mathbf{1} + c\,P_{2,M}\big) - \sqrt{\Omega}\,\big(N + \tfrac{1}{2}\big)\Big|_M, \qquad c = \frac{1}{\gamma\Omega},$$

with $P_{2,M}$ the truncated momentum of mode 2 (S_M is the channel operator of Proposition 10.7 taken with the standard overall sign; $A \mapsto -A$ interchanges the deficiency indices, and S_M commutes with the natural conjugation of its decoupled eigenframe (equivalently $K \circ \Pi_2$ in the mode-2 Fock basis), so $n_+ = n_-$ and the overall sign is immaterial), has deficiency indices $(2M, 2M)$.

Proof. $Q(k\mathbf{1} + cP_{2,M})$ is a polynomial in the single Hermitian matrix $P_{2,M}$, hence exactly diagonal in its constant (k-independent) eigenframe; the eigenvalues p_ν of $P_{2,M}$ (scaled Gauss–Hermite points) are simple. In that frame S_M is a direct sum of the scalar operators $\hat{p}^2/(2\gamma\Omega) - Q(k + c\,p_\nu)$—each limit circle at both ends [23], contributing $(2, 2)$—plus the rotated channel-energy matrix, a *constant bounded* Hermitian perturbation, which preserves deficiency indices [25]. □

The falsifiable consequence—with couplings on, the wall-regularized level density must equal the all-$2M$-branch sum $\sum_\nu T_{\mathrm{cross},\nu}/\pi$; were any channel's deficiency destroyed, the density would drop—is confirmed numerically for $M = 1, 2, 3$ at three wall positions (count-based spacing matching the prediction within count granularity; the exact frame decoupling holds to 10^{-13},

and the $\pm p_\nu$ branch ladders visibly lock into parity doublets). What remains for the full operator is only the channel tail: the exact Fock compression differs from the Galerkin model by terms supported on the top four channels, and control of the deficiency vectors' decay in channel number as $M \to \infty$ is open—equivalently, the full operator is a direct integral of limit-circle fibers over $\mathrm{spec}(p_2)$ coupled across fibers by the transverse oscillator.

Corollary 10.9 (sign dichotomy at every finite channel number)**.** *For $\lambda < 0$ the same frame decoupling yields the confining branches $\hat{p}^2/(2\gamma\Omega)+|\lambda|(k+c\,p_\nu)^4$ (plus exact lower-order quadratic terms, not a remainder), each essentially self-adjoint; with the bounded rotated channel-energy term (Kato–Rellich), every finite-channel Galerkin model of the $\lambda < 0$ quartic PU is essentially self-adjoint. Thus at every M the deficiency indices are $(2M, 2M)$ for $\lambda > 0$ and $(0, 0)$ for $\lambda < 0$.*

Numerically, the $\lambda < 0$ wall-regularized spectra converge as the wall recedes (matched drift 3×10^{-8}, then 10^{-13})—the regulator-independence fingerprint, in maximal contrast to the never-converging $\lambda > 0$ sweep.

Mechanisms of the two completion candidates. Tracking the actual classical blowup trajectories and evaluating the kinetic form on the escaping momentum discriminates the mechanisms sharply. The $\lambda > 0$ PU (the deficiency case) escapes along a *definite* kinetic direction (form ratio $\to 0.9997$). The ghost pair's escape repeatedly returns to the *null* diagonal valley: at the four sampled amplitude-threshold crossings the kinetic-form ratio $(p_1^2 - p_2^2)/(p_1^2 + p_2^2)$ is small (0.01–0.07), but this is not a trajectory-wide bound—between crossings the ratio oscillates across the full range—so the escape never locks onto a definite kinetic direction the way the $\lambda > 0$ PU does ($\to 0.9997$), supporting no sustained definite-kinetic WKB transport, a structural rationale for its quantum completion. We conjectured that completion is equivalent to null escape; the $\lambda < 0$ PU *falsifies* the naive form of this conjecture: its ratio also tends to 1. Inspection shows it climbs a *rising* potential using negative kinetic energy while hugging the line $u = -cv$—that is, it escapes through arbitrarily high mode-2 channels, precisely the channel tail excluded by the Galerkin models. The two candidates therefore realize two distinct completion mechanisms (null-direction escape versus channel-tail escape), the second more tail-sensitive than the first. Finally, the ghost-pair verdict is robust under valley-adapted bases: in rotated (s, d) Fock bases aligned with the escape diagonal, at three squeezing choices including strongly valley-penetrating ones, the $\lambda < 0$ drift ratios (0.16–0.45) interleave with the free ($\lambda = 0$) control's range—nothing approaching the limit-circle band.

Finally, the $M \to \infty$ tail itself.

Proposition 10.10 (direct-integral deficiency; the exact form of the tail question)**.** *The oscillator-free part $A_0 = \hat{p}_k^2/(2\gamma\Omega) - Q(k+c\,p_2)$ commutes with p_2: it is a direct* integral *over* $\mathrm{spec}(p_2) = \mathbb{R}$ *of shifted scalar fibers, each limit circle for $\lambda > 0$, and measurable fields of fiber deficiency solutions give $n_\pm(A_0) = \infty$. The full channelized operator differs from A_0 by the transverse oscillator $h_2 = p^2/(2\gamma) - (\gamma\Omega/2)\partial_p^2$, which in this exact frame is* unbounded *(its Galerkin compressions have norm $\sqrt{\Omega}(M - \frac{1}{2}) \to \infty$)—precisely why the finite-channel stability theorem (Theorem 10.8) does not pass trivially to $M \to \infty$.*

Remark 10.11 (no compact witnesses). For any symmetric operator defined on C_0^∞ and any $\mathrm{Im}\, z \neq 0$, every compactly supported smooth ψ lies in the closure's domain, where $\|(S - z)\psi\| \geq |\mathrm{Im}\, z|\ \|\psi\|$ holds unconditionally. *No compactly supported—windowed or lattice-regularized—function can witness deficiency, nor any finite basis (Fock/Hermite) truncation, which lies in the closure's domain by a cutoff argument (the operators here have polynomially bounded coefficients)*: the completion data lives entirely at infinity, in the same sense in which the completions

themselves are an S-matrix at infinity (Theorem 10.2). All truncation-based diagnostics, including ours, are therefore evidence in principle, never proof; genuine von Neumann certificates require candidates with analytically controlled tails.

Such a certificate is implementable in one dimension: gluing WKB tails (whose only residual is the explicit remainder $a[q''/(4q) - \frac{5}{16}(q'/q)^2]$, $q = (z-U)/a$ with U the fiber potential and a its leading coefficient) to an exactly integrated interior yields, for the falling-quartic fiber, a machine-checked ratio $\|(S - \mathrm{i}Z)\psi\| / (Z\,\|\psi\|) = 0.0011 \ll 1$: the 1D fiber operator is *certified*—in the computer-assisted sense: the residual is evaluated numerically with a three-orders safety margin, not rigorously error-bounded—not essentially self-adjoint. The construction is intrinsically sharp: for the confining control the left continuation contains a non-L^2 real-exponential branch (no candidate exists), and for the marginal falling-quadratic case the growing branch is polynomial, $|\psi| \sim k^{-1/2+Z/1.12}$, non-$L^2$ for every $Z > 0$—consistent with the limit-point verdict of Theorem 8.1(1). On the full two-dimensional operator, windowed candidates supply a *closure-defect diagnostic*: falling-channel operators hug the forbidden bound (ratios 1.21–1.41 for the five $\lambda > 0$ beam launches—four at 1.21–1.29, the outer-window launch at 1.41—bracketing the 1D limit-circle control at 1.22), while the $\lambda < 0$ control sits at 641 and the 1D confining control at 4.9. The remaining mathematical content of the $M \to \infty$ question is thereby isolated: a two-dimensional certificate requires analytic beam tails along the escape rays.

A Maslov feasibility study (`scripts/maslov_feasibility.py`) indicates the geometry of those tails. In the null-frame coordinates the ray equations of the symbol reduce *exactly* to the quartic-PU law $s'''' + \Omega s'' + \Omega_0^2 s = (4\lambda/\gamma)\, s^3$, so the escape-ray family is the classical blowup itself, with universal asymptotics $s \simeq A\,(t_* - t)^{-2}$, $A = \sqrt{30\gamma/\lambda}$: measured exponent -1.999 and amplitude 24.55 against $\sqrt{600} = 24.49$, at every launch point tested, in *both* time directions—rest data escapes at both temporal ends, so the time-integral quasimode $\psi = \int \mathrm{e}^{\mathrm{i}zt}\varphi_t\, dt$ over a doubly-explosive ray has only $\mathrm{e}^{-Zt_*} \approx 10^{-14}$ boundary terms. Along these rays the Gaussian-beam data [59] are favorable precisely where analytic tails are needed: Siegel positivity of the complex curvature holds throughout, its numerical-zero direction aligned with the flow (alignment 1.0000; benign, integrated over in t); the transverse curvature stays positive with strong terminal focusing (width 5×10^{-3} at the escape end); no caustics occur; $|\det \partial q| \sim \tau^{-3.1}$ makes the beam-norm integral converge ($N = 0.082$ at $Z = 6$); and the anharmonic residual proxy decays rapidly along the escape (weighted integral 0.026), failing only in a *compact interior episode* of transverse defocusing (peak 8×10^3)—exactly the region the one-dimensional template already handles by exact numerical solution. The two-dimensional glued certificate thus should reduce to: numeric interior on a compact region, Gaussian-beam tails along the known rays, gluing where the residual proxy is small—assembly and quantitative error control rather than new asymptotic machinery.

Part III
Contact with the world

11 Experimental probes

The closed forms of this paper are testable with existing devices; we collect the concrete predictions.

(a) The terminal-indivisibility theorem in a lossy photonic coupler. The dimer (4) is the

standard system of experimental $\mathcal{PT}$ optics (coupled waveguides or resonators with engineered loss), and the correspondence's objects are normalized intensities, so classical light suffices. The apparatus exists: passive $\mathcal{PT}$ lossy couplers are characterized as functions of propagation length [50], information-flow criticality at the exceptional point has been measured in single-photon dimers [51], and divisibility *classes* (P- vs CP-divisibility) are experimentally resolvable [52]. What has not been extracted is precisely this paper's observable set: a closed-form divisibility-onset length, a generator floor, and the recurrent-window period. Protocol: measure the intensity transfer matrix $\Gamma(t)$ at a sequence of propagation lengths, form the divisor $M = \Gamma(t_2)\Gamma(t_1)^{-1}$ (2), and locate the onset of negativity. Theorem 4.1 predicts, with no free parameters beyond the device constants (θ, s): onset at $t_c = \tilde{\kappa}^{-1}\mathrm{arcsinh}(\tilde{\kappa}/(s\sqrt{2}))$, *permanent* thereafter in the broken phase, with the generator floor $-(\sin\theta + \tilde{\kappa})$; *recurrent* windows of period π/κ in the unbroken phase; and t_c continuous through the exceptional point. The dissociation of Sec. 4 is testable on the same data set by running the two analyses (trace-distance backflow versus divisibility) side by side: backflow dies at the exceptional point while the indivisibility onset crosses it smoothly.

(b) The conditioning signal on an entangled pair. Applying the local lossy dimer to one arm of an entangled photon pair and post-selecting on survival shifts the partner's marginal by a closed-form amount with late-time magnitude $\sqrt{1-(s/\sin\theta)^2}$—zero at the exceptional point, approaching unity deep in the broken phase—while the *unconditioned* marginal is exactly unchanged (derivation in the companion sequel; the closed form follows from Theorem 4.1's p, q, r). This is a direct laboratory test of the sample-space bookkeeping that this paper argues is the content of broken-$\mathcal{PT}$ physics, and it locates the apparent no-signaling violation of local $\mathcal{PT}$ operations reported in the literature in the post-selection, quantitatively. The entangled-pair-plus-tailored-loss platform exists [53, 54]; the closed-form magnitude is the new target.

(c) The skin obstruction in circuits and lattices. Hatano–Nelson physics is routinely realized in topolectrical circuits and photonic lattices; Theorem 3.4 identifies the measured skin envelope with the beable metric, $d_i = \mathrm{e}^{2wi}$, and Corollary 3.5 predicts a specific new experiment: two opposite-field chains coupled by reciprocal rungs (the $\mathbb{Z}_2$-type ladder, net winding zero) are obstructed at *any* rung coupling—observable as irreducible nonreciprocal transport that no local gauge transformation removes. The $\mathbb{Z}_2$ skin effect itself has recently been observed (acoustically and in programmable circuits [55]); the prediction here is not its existence but the *beable-obstruction observable*: the irreducibility of the nonreciprocity, equivalently the detailed-balance violation of the underlying rate graph, which mode-localization measurements do not extract.

(d) The explosion dictionary in engineered potentials. Level-spacing *saturation*, $\Delta E = \pi/T_{\mathrm{cross}}$ approaching a finite limit as the box grows rather than decaying to zero (Table 2), is the spectroscopic fingerprint of reflection off infinity at finite recurrence time (the underlying boundary-condition spectra are classical mathematical physics [56]; the saturation framing as an observable appears new); engineered super-quadratically falling potentials in cold-atom or circuit-QED simulators could exhibit it directly—inverted-oscillator BEC platforms with full state tomography now exist [57] and are one nonlinearity away from the $-x^4$ class. On the transient side, the cascade laws (14) are two-mode squeezing, already measured as vacuum pair creation in the dynamical Casimir effect [58]; engineered superlinearly Bose-enhanced ladders would probe the transient-to-explosive transition itself.

12 Discussion

12.1 What each program learns

For the antilinearity program. Closed-system $\mathcal{PT}$ machinery adds no new stochastic processes: in the unbroken phase it is an isomorphism onto ordinary quantum mechanics (the deflation proposition, Proposition 2.1), at the EP its coordinates degenerate (Sec. 5), and its genuinely new regime—the broken phase—is most naturally read as *open-system* physics: survival-conditioned dynamics whose stochastic signature is the recurrent $\to$ terminal transition of indivisibility (Theorem 4.1). For PU, the no-ghost construction is domain data at infinity rather than dynamics on the configuration space (Sec. 6); the explosion dictionary (Theorem 8.1(2)(a)) and the channel-additivity theorem (Theorem 10.2) show precisely what such data amounts to.

For the stochastic–quantum correspondence. Antilinear symmetry is the exact existence gate for closed processes; detailed balance (the detailed-balance theorem, Theorem 3.1) is the exact fixed-beable gate, resolving both conceptual costs on that class and giving the metric weights a classical meaning (inverse stationary distribution). The correspondence handles ghosts gracefully—too gracefully to see them at free level: the ghost enters the stochastic currency only through recurrence classification (conservative $\to$ transient $\to$ explosive), and the minimal explosive process is the only completion-independent object.

12.2 The trilemma

For a malicious interacting ghost one must choose: (i) accept the minimal explosive process—substochastic, the dilation structure of Sec. 4, canonical; (ii) buy conservativity with an S-matrix at infinity—in the separable multi-D model an infinite-dimensional, channel-scrambling reflection law injected by hand (Theorem 10.2); or (iii) repair the spectrum by moving the domain—which exits the sample space (Sec. 6). In Barandes' currency, (i) is the answer; vacuum decay *is* explosion in the probabilist's standard sense.

12.3 Outlook

Open directions: Open Problem 10.6, whose two-dimensional certificate should reduce to assembly and error control over the beam data supplied in Sec. 10; field theory, where the explosion/transience distinction should meet vacuum decay rates; the interacting broken phase, where terminal indivisibility and transience could combine; and extensions of the skin-effect correspondence of Sec. 3.1 to higher-dimensional and symmetry-protected skin effects [48], where the obstruction should acquire the corresponding topological classification.

Author contributions and use of AI

This manuscript was produced through a collaboration, directed by the author, with the large language model Claude Fable 5 (Anthropic). The AI system carried out essentially all of the technical work: the research program and its strategy, the analytical derivations and proofs, the design, implementation, and interpretation of all numerical experiments and figures, the literature search and reference verification, and the drafting and revision of the manuscript, including this statement. J.T. posed the research direction, provided iterative review and quality feedback, and takes sole responsibility for the decision to circulate the manuscript. In line with prevailing editorial policies (e.g., COPE guidance and the policies of the major physics

publishers), the AI system is not listed as an author, since authorship entails accountability that an AI system cannot hold; this statement is intended to make the provenance unambiguous. Every numerical claim is reproducible from the accompanying scripts, and every theorem-level claim is presented with its proof, so that all results can be verified independently of their origin.

Code availability

All scripts, the running lab notes, and the LaTeX source of this manuscript are publicly available at

https://github.com/embeddetech/p01-indivisible-ghosts

The repository also preserves the provenance record: the original conversation in which the research question was posed (`Conversation_Conformal_Gravity_PT_Barandes.pdf`) and the project briefing that seeded the working sessions (`PT_stochastic_handoff.md`). All results are reproducible from short standalone Python scripts (NumPy/SciPy); most run in seconds to minutes on a laptop, while the two long trajectory-integration scripts (`scripts/interacting_pu.py`, `scripts/completion_candidates.py`) can take tens of minutes, machine- and SciPy-version-dependent:

- Sec. 2: `scripts/pt_barandes.py`
- Sec. 3: `scripts/fixed_beable_kolmogorov.py`, `scripts/skin_effect_beables.py`, `scripts/skin_topology_beables.py`
- Sec. 4: `scripts/dilation_bridge.py`, `scripts/lemma_terminal_indivisibility.py`
- Sec. 5: `scripts/exceptional_point.py`
- Sec. 6: `scripts/pais_uhlenbeck.py`
- Sec. 7: `scripts/interacting_pu.py`, `scripts/quartic_pu_leak.py`
- Sec. 8: `scripts/explosion_theorem.py`
- Sec. 10: `scripts/deficiency_multiD.py`, `scripts/pu_deficiency_evidence.py`, `scripts/pu_bo_channels.py`, `scripts/gap_closure_channels.py`, `scripts/completion_candidates.py`, `scripts/tail_certificate.py`, `scripts/glued_certificate_1d.py`, `scripts/maslov_feasibility.py`
- App. B (escape dictionary): `scripts/appD_escape_dictionary.py` (central-potential T_ℓ ratios and the shelf-potential per-channel T_m)
- Figures: `scripts/make_figures.py` regenerates all figures except Fig. 1, which is produced by `scripts/skin_effect_beables.py`

A Proof of the escape-diffusion lemma

This appendix proves Lemma 8.2 (the stochastic leg of Theorem 8.1): for the escape diffusion $\mathrm{d}X = v(X) \circ (\mathrm{d}t + \varepsilon\, \mathrm{d}W)$ (or its driftless variant), $v = \sqrt{2(E - V)}$, reflected at x_0 and minimal at $+\infty$, the boundary $+\infty$ is regular iff $T(E) < \infty$ and natural iff $T(E) = \infty$, and the minimal semigroup is sub-stochastic (explosive) in the regular case.

Transport to the clock coordinate $\tau : [x_0, \infty) \to [0, T)$, a C^1 increasing bijection ($v \geq v_{\min} = \sqrt{2(E - \sup V)} > 0$); scale and speed are intrinsic under such maps, and the images of the two diffusions are Brownian motion on $[0, T)$ with drift $\beta \in \{0, 1\}$ and variance ε^2: $\mathrm{d}\tilde{s} = \mathrm{e}^{-\varrho u}\mathrm{d}u$, $\tilde{m}(\mathrm{d}u) = \frac{2}{\varepsilon^2}\mathrm{e}^{\varrho u}\mathrm{d}u$, $\varrho = 2\beta/\varepsilon^2$. With $\Sigma = \int^\infty M\,\mathrm{d}S$ and $N = \int^\infty S\,\mathrm{d}M$ (boundary classification of [27, 28]; [30], Ch. 15.6; [29]): for $T < \infty$ both $\tilde{S}(0, T)$ and $\tilde{M}(0, T)$ are finite, so $\Sigma, N \leq \tilde{S}\,\tilde{M} < \infty$ and $+\infty$ is regular; the driftless case (unit noise $\varepsilon = 1$) has $\tilde{S}(0, \xi] = \xi$, $\tilde{M}(0, \xi] = 2\xi$, whence $\Sigma = \int_0^T 2\xi\,\mathrm{d}\xi = T^2 = N$ (a general $\varepsilon > 0$ rescales Σ, N by ε^{-2} and leaves the regular/natural dichotomy unchanged). For $T = \infty$ both diverge, so $+\infty$ is natural; $\Sigma < \infty \Leftrightarrow T < \infty \Leftrightarrow N < \infty$, so exit and entrance never occur, proving (i); E-independence follows from $(E - V)/(E' - V) \to 1$ as $x \to \infty$.

For (ii) with $T < \infty$, the reflected process hits T almost surely (Skorokhod: $Z_t \geq z + \beta t + \varepsilon W_t \to \infty$; reflecting Brownian motion is recurrent when $\beta = 0$), so $\zeta < \infty$ a.s., and as T is accessible (reachable with positive probability), $P_t^{\min}\mathbf{1}(x) = \mathbb{P}_x(\zeta > t) < 1$. Solving $D_m D_s u = -1$ with $D_s u(x_0) = 0$ and $u(\infty) = 0$ gives $\mathbb{E}_{x_0}[\zeta] = \Sigma$ ([30], §15.3); the small-noise sandwich $t + \varepsilon W_t \leq Z_t \leq t + 2\varepsilon \sup_{s \leq t} |W_s|$ forces $\zeta \to T$ a.s. as $\varepsilon \downarrow 0$ ($\beta = 1$). For $T = \infty$, $\Sigma = \infty$ makes $+\infty$ inaccessible (Feller's explosion test), so $\zeta = \infty$ a.s. and $P_t^{\min}\mathbf{1} = \mathbf{1}$; uniqueness at a natural boundary is Feller's theorem, no lateral condition being admissible. Part (iii) is the classification of lateral conditions at a regular boundary [28, 29], matched to (2)(a)–(b). Finally, accessibility forces $S < \infty$, whence $N \leq S \cdot M < \infty$ whenever the speed mass is finite: an accessible boundary honoring the WKB identity is regular, never exit.

B The multidimensional escape dictionary

This appendix extends the one-dimensional escape-diffusion leg of the explosion dictionary (Theorem 8.1(2)(c)–(d), Lemma 8.2) to the channel setting of Sec. 10, providing the exact stochastic mirror of the channel-additivity and bounded-mixing theorems (Theorems 10.2 and 10.4) and locating precisely where the dictionary stops being a theorem. The verdict is *per-channel-only*: the dictionary generalizes verbatim channel by channel in the separable, central, and boundedly mixed cases, and the genuine obstruction is unbounded channel transfer—whose two failure modes turn out to be asymmetric in exactly the way the paper's numerical verdicts are (conservativity robust, explosivity fragile; Theorem B.11).

Status conventions. Every statement below is tagged *[proved]*, *[proved, sketch]* (rigorous reduction to cited standard machinery, details routine but not written out), or *[conjecture]*. Lemma 8.2 is used throughout in the form stated in Sec. 8 and proved in Appendix A: for the escape diffusion $\mathrm{d}X = v(X) \circ (\beta\,\mathrm{d}t + \varepsilon\,\mathrm{d}W)$, $v = \sqrt{2(E - V)}$, $\beta \in \{0, 1\}$, reflected at the base point and minimal at $+\infty$, the boundary $+\infty$ is *regular* iff the classical time of flight $T = \int^\infty \mathrm{d}x/v$ is finite (minimal semigroup sub-stochastic and explosive) and *natural* iff $T = \infty$ (conservative, unique Feller realization), the driftless speed measure being twice the WKB deficiency density of (16).

B.1 What replaces the single end and the single integral

In one dimension the escaping end is a point and one integral $T(E)$ decides all three legs. In the separable setting of Theorem 10.2,

$$H \;=\; h_1 \otimes \mathbf{1} + \mathbf{1} \otimes h_\perp \quad \text{on } L^2(\mathbb{R} \times \mathbb{R}^m), \qquad h_1 = \tfrac{1}{2}p^2 + V_1, \quad h_\perp \varphi_n = E_n \varphi_n,$$

the classical escape mechanism conserves the transverse energy and the quantum deficiency vectors are $u_n^{\pm} \otimes \varphi_n$. The correct multidimensional replacements, established below, are:

- the single escaping end $+\infty$ becomes the *countable set of channel ends* $\{\infty_n\}_{n\in\mathbb{N}}$, one per transverse eigenstate (for continuous transverse spectrum, a measure space of fiber ends; Sec. B.6);
- the time of flight becomes the *per-channel* family

$$T_n \;=\; T(E-E_n) \;=\; \int_{x_n}^{\infty} \frac{\mathrm{d}x}{\sqrt{2\,(E-E_n-V_1(x))}},$$
$$x_n \text{ any base point with } \sup_{x\geq x_n} V_1 < E-E_n;$$

  this is the precise form taken by a would-be direction-dependent $T(E,\omega)$: the direction label ω is quantized into the channel label n (for central potentials, literally the angular decomposition, Corollary B.5);
- and—genuinely new in $d>1$—a *second* kind of end, the *channel tail* $n\to\infty$ at finite or co-moving x, with no 1D counterpart, which is exactly where the clean dictionary stops being a theorem (Secs. B.5–B.6).

Throughout, V_1 is continuous with $V_1(x)\to-\infty$; where the quantum leg is invoked we assume in addition the regularity of Theorem 8.1, including its added C^2 control ($|V''|\,|V|^{-3/2}$ integrable, Prop. C.4). A first elementary but structurally important fact:

Lemma B.1 (channel class equality; proved)**.** *If $V_1(x)\to-\infty$ then for all n,m: $T_n<\infty \iff T_m<\infty$. Hence the* qualitative *classification is channel-uniform; only the* quantitative *data $(T_n)_n$ is channel-dependent.*

Proof. $\big(E-E_n-V_1(x)\big)/\big(E-E_m-V_1(x)\big)\to 1$ as $x\to\infty$, so the integrands are comparable near $+\infty$; the finite parts are finite by continuity and the choice of base points. □

This is the stochastic shadow of the λ-independence of the limit-point/limit-circle dichotomy used in the proof of Theorem 10.2: there, every channel is limit circle because one channel is; here, every channel end is regular because one is.

B.2 The separable case: full per-channel generalization

Define the state space as the disjoint union of half-lines $\mathcal{E}=\bigsqcup_{n\in\mathbb{N}} I_n$, $I_n=[x_n,\infty)$, and on each I_n the channel escape diffusion $X^{(n)}$ of Lemma 8.2 with speed $v_n(x)=\sqrt{2(E-E_n-V_1(x))}$, clock $\tau_n(x)=\int_{x_n}^{x}\mathrm{d}y/v_n(y)$, reflection at x_n, minimal (absorbing) convention at ∞_n; generator $\mathcal{L}=\bigoplus_n L_n$, $L_n=\frac{\varepsilon^2}{2}(v_n\partial_x)^2+\beta\,v_n\partial_x$, $\beta\in\{0,1\}$. This is the stochastic realization of the separable mechanism: the transverse motion is bounded oscillation whose only dynamical residue is its conserved energy, i.e. the frozen channel label.

Theorem B.2 (channel dictionary, separable case; proved except the countable part of (iv))**.** *Let V_1 be continuous, $V_1\to-\infty$, and let (E_n,φ_n) be as above. Then:*

(i) ***(per-channel classification)*** *Each channel end ∞_n is* regular *iff $T_n < \infty$ and* natural *iff $T_n = \infty$ (never exit, never entrance); by Lemma B.1 all channel ends are in the same class. For $\beta = 0$ (unit noise, $\varepsilon = 1$) the Feller test integrals in channel n are $\Sigma_n = N_n = T_n^2$ (in general T_n^2/ε^2).*

(ii) ***(explosion)*** *If $T_n < \infty$ (equivalently, for all n): started in channel n, the process explodes a.s. with mean lifetime T_n^2 ($\beta = 0$, $\varepsilon = 1$) resp. $T_n - \frac{\varepsilon^2}{2}\big(1 - \mathrm{e}^{-2T_n/\varepsilon^2}\big)$ ($\beta = 1$), and $\zeta \to T_n$ a.s. as $\varepsilon \downarrow 0$ ($\beta = 1$):* each channel loses its mass at its own classical escape time. *The minimal semigroup $\bigoplus_n P_t^{\min,n}$ is strictly sub-stochastic on every channel. If $T_n = \infty$ for one (hence every) n, the direct sum is conservative and is the unique Feller realization of $\mathcal{L}$ with the given reflecting data.*

(iii) ***(deficiency = speed-measure data, channel by channel)*** *The driftless speed measure of channel n is the WKB density (16) of the deficiency vector $u_n \otimes \varphi_n$ of Theorem 10.2: $m_n(\mathrm{d}x) = 2[2(E - E_n - V_1(x))]^{-1/2}\mathrm{d}x \approx 2\,|u_n^{\mathrm{WKB}}(x)|^2\,\mathrm{d}x$, with scale mass T_n near ∞_n. Thus deficiency indices (∞, ∞) correspond exactly to* countably many boundary ends, each carrying finite scale and speed mass*—one Feller datum per deficiency channel.*

(iv) ***(lateral conditions = the S-matrix at infinity)*** *Suppose $T_n < \infty$ for all n, and truncate to M channels. The sub-Markov Feller extensions of the minimal process are exactly the Feller–Wentzell lateral conditions at the M regular ends: at each ∞_n a convex datum (absorption, killing, reflection, stickiness, and a jump-in measure ν_n on $\mathcal{E}$), the jump-in measures allowed to charge* other *channels. The conservative extensions are those with no absorption/killing; a packet escaping in channel n then re-enters distributed according to ν_n—a channel-*mixing *return law, the stochastic mirror of the unitary S-matrix at infinity of Theorem 10.2, and different lateral data reproduce the regulator-dependence of the returned mixture seen in Fig. 8.* [Finite M: proved, via Feller's lateral conditions and Freidlin–Wentzell gluing; countable M: proved, sketch, via Itô excursion theory with a summability condition on the excursion entrance laws; the completeness claim—that all Feller extensions arise this way—is standard for finite M and conjectural for countable M.]

Proof. (i)–(iii) are channelwise reductions. The direct sum has no interaction between channels, so every path quantity is computed in the channel where the process starts; Lemma 8.2 applied with potential $V_1 + E_n$ (spectral shift $E \mapsto E - E_n$) gives (i) and (ii) verbatim, and Lemma B.1 gives the class equality. Uniqueness in (ii): at a natural end no lateral condition exists (Feller [27, 28]), channel by channel, and a Feller extension whose generator agrees with the direct sum on $C_c(\bigsqcup_n I_n^\circ)$ respects the direct-sum structure away from the boundary (no interior channel-coupling jumps); since no boundary is accessible, the minimal realization is the only one. For (iii), the deficiency equation in channel n is $(h_1 - (\pm\mathrm{i} - E_n))u = 0$; to leading WKB order the modulus is $|u_n|^2 \approx [2(E - E_n - V_1)]^{-1/2} = \frac{1}{2}m_n$, and the scale mass is $\int^\infty \mathrm{d}\tau_n = T_n$.

(iv), finite M. Each ∞_n is regular by (i). Feller's classification of lateral conditions at a regular boundary of a generalized second-order operator [28] provides, for one interval, the general boundary datum including a jump-in measure; the multi-interval (graph) case, with boundary points allowed to communicate, is Freidlin–Wentzell gluing (Freidlin and Wentzell, Ann. Probab. **21** (1993) 2215), which parametrizes all Feller processes agreeing with the minimal one in the interior by Wentzell data at the vertices; jump-in to interior points is supplied by the Itô excursion construction of instantaneous-return processes (Itô, Proc. Sixth Berkeley Symp. III

(1972) 225; Blumenthal, *Excursions of Markov Processes*, Birkhäuser 1992). Countable M: the same construction applies end by end; one must check that concatenation over countably many ends accumulates no infinitely many boundary visits in finite time, for which a uniform lower bound on mean return times (automatic when $\inf_n T_n > 0$, in general a summability condition on the entrance laws) suffices. We assert completeness only for finite M. □

Remark B.3 (the mirror is structural, not a bijection). The quantum completions form the unitary group $U(\mathcal{K}_+, \mathcal{K}_-)$, $\dim \mathcal{K}_\pm = \infty$ (Theorem 10.2); the conservative stochastic completions form the convex set of Markov gluing data (ν_n). As already in one dimension ($U(1)$ versus Feller's convex family, Lemma 8.2(iii)), these are *parallel infinite-dimensional families attached to the same boundary objects*, in canonical correspondence at the level of which channels communicate through infinity, not isomorphic as spaces.

Remark B.4 (conservative / transient / explosive trichotomy). The hierarchy of Sec. 7 is reproduced by the clock picture: in channel n the drifted model is Brownian motion with unit drift on $[0, T_n)$; $T_n = \infty$ with $\beta = 1$ is *conservative and transient* (the quadratic-ghost cascade, Theorem 10.1), $T_n < \infty$ is *explosive*, and a confining channel (no escape interval) is *recurrent*.

B.3 Central potentials: the sphere at infinity, resolved into channels

Corollary B.5 (central case; proved)**.** *Let $H = -\frac{1}{2}\Delta - c|x|^\alpha$ on $\mathbb{R}^d$, $d \geq 2$, $c > 0$, $\alpha > 2$ (Corollary 10.3). In the angular decomposition the radial channel ℓ carries the escape diffusion with $v_\ell(r)^2 = 2(E + cr^\alpha - L^2/(2r^2))$, $L^2 = (\ell + \frac{d-2}{2})^2 - \frac{1}{4}$, on $r \geq r_\ell$ beyond the outer turning point. Then every channel end is regular, with*

$$T_\ell \asymp C_\alpha\, c^{-\frac{2}{\alpha+2}}\, L^{\frac{2-\alpha}{2+\alpha}} \xrightarrow[\ell\to\infty]{} 0,$$

so $T^* = \sup_\ell T_\ell < \infty$: the classification is uniform across the entire sphere of directions. All conclusions of Theorem B.2 hold with $n \mapsto \ell$; in particular the conservative completions of the central malicious ghost are angular-momentum-mixing return laws at $r = \infty$.

Proof. Substituting $r = \lambda\rho$ with $\lambda = (L^2/c)^{1/(\alpha+2)}$ equalizes the potential and centrifugal scales, giving $T_\ell = c^{-1/2}\lambda^{1-\alpha/2}(C_\alpha + o(1))$ with $C_\alpha = \int_{\rho_0}^\infty (2(\rho^\alpha - \frac{1}{2}\rho^{-2}))^{-1/2} \mathrm{d}\rho < \infty$ (convergent at infinity since $\alpha > 2$, at the simple turning point as $(\rho - \rho_0)^{-1/2}$). Numerical check ($\alpha = 4$, $c = 1$, $E = 0$; `scripts/appD_escape_dictionary.py`): $C_\alpha = 0.964$ and T_ℓ at $L^2 = 10^2, 10^4, 10^6$ is $0.447, 0.208, 0.096$, successive ratios 2.154 against the predicted $10^{1/3} = 2.154$. The rest is Theorem B.2. □

Thus for central potentials the anticipated per-direction $T(E, \omega)$ collapses to the channel family T_ℓ: the boundary at infinity is not the geometric sphere S^{d-1} but its harmonic resolution, and the uniformity $T^* < \infty$ is automatic—the multidimensional situation in which the generalization is *complete*.

B.4 Bounded channel mixing: the stochastic Kato theorem

Theorem 10.4 says a bounded mixing preserves the deficiency indices. The stochastic counterpart replaces the frozen channel label by a jump process with *bounded intensity*. Fix measurable rates $q(x; n \to m) \geq 0$ with $\sup_{x,n} \sum_m q(x; n \to m) \leq \bar{q} < \infty$, supported on open channels; realize (X_t, n_t) by thinning an intensity-$\bar{q}$ Poisson clock, the path solving the frozen channel SDE between accepted switches and continuous across them.

Theorem B.6 (stability under bounded mixing; proved)**.** *For the switched escape process:* ***(a)*** *if $T_n = \infty$ for one (hence every) channel, the process is conservative for* every *admissible bounded-rate law;* ***(b)*** *if* (U) : $T^* := \sup_n T_n < \infty$*, then it explodes a.s. for every admissible law, with* $\mathbb{E}[\zeta] \le h/p_* < \infty$ *where* $h = T^* + 1$ *and* $p_* = \mathrm{e}^{-\bar{q}h}\mathbb{P}(\varepsilon \sup_{s\le h}|W_s| \le \frac{1}{2}) > 0$ *($\beta = 1$; the analogous positive constant for $\beta = 0$). Hence under (U) the dictionary survives bounded mixing verbatim—the stochastic mirror of Kato stability of the deficiency indices (Theorem 10.4, [25]).*

Proof. (a) The candidate switch times are Poisson($\bar{q}$), so a.s. finitely many switches in $[0, t]$; if $\zeta \le t$ the explosion instant lies in an inter-candidate interval on which the path is a frozen channel diffusion with natural end (Lemma 8.2, $T = \infty$), inaccessible in finite time—a null event, and a countable union of null events is null. (b) On windows $[kh, (k+1)h)$ let A_k be the event of no candidate in the window and $\varepsilon \sup_{s\le h}|W_{kh+s} - W_{kh}| \le \frac{1}{2}$; $\mathbb{P}(A_k \mid \mathcal{F}_{kh}) \ge p_*$. On A_k, in the clock coordinate (proof of Lemma 8.2) $Z_s \ge s - \frac{1}{2}$, and $s = T_n + \frac{1}{2} \le T^* + \frac{1}{2} < h$ forces Z to reach T_n: explosion inside the window. Thus $\mathbb{P}(\zeta > (k+1)h \mid \zeta > kh) \le 1 - p_*$, giving the geometric bound. The argument never needs which channels are visited—only (U) and open-channel support. □

Lemma B.7 (when (U) is automatic; proved)**.** *If for $x \ge R$: $-V_1(x) \ge c_1 x^{2+\epsilon}$ and $-V_1'(x) \ge c_3 x^{1+\epsilon}$ ($\epsilon > 0$, $V_1 \in C^1$ strictly decreasing), and $T(E) < \infty$, then $\sup_{E'\le E} T(E') < \infty$; so (U) holds for every transverse spectrum. This covers all power-law potentials $-V_1 \sim cx^{2+\epsilon}$, hence the paper's representative models.*

Proof. With $\mu = -E'$ and turning point x' ($-V_1(x') = \mu$, so $x' \le x_g := (\mu/c_1)^{1/(2+\epsilon)}$ and $x' \to \infty$ as $\mu \to \infty$), integrate the lower bound $-V_1' \ge c_3 y^{1+\epsilon}$ over $[x', x]$: $-\mu - V_1(x) \ge \frac{c_3}{2+\epsilon}(x^{2+\epsilon} - x'^{2+\epsilon})$. Substituting $x = x'w$, $T(E') = \int_{x'}^{\infty} \mathrm{d}x/\sqrt{2(-\mu - V_1)} \le C\, x'^{-\epsilon/2} \int_1^{\infty} \mathrm{d}w/\sqrt{w^{2+\epsilon} - 1} = C'' x'^{-\epsilon/2} \to 0$ (the w-integral converges); so $T(E') \to 0$ as $E' \to -\infty$, and T is finite and continuous on compact E'-ranges (simple turning points). □

B.5 The obstruction: unbounded channel transfer

Everything so far generalizes. What follows is the exact point of failure, proved by construction.

Lemma B.8 (channel-riding reduction; proved)**.** *Let $n(\cdot) : [x_*, \infty) \to \mathbb{N}$ be measurable, locally constant off a discrete set, with $n(x)$ open at x. The* position-locked *switched process (switch to $n(x)$ upon reaching x; a limit of bounded-rate laws with rates $\to \infty$,* outside *Theorem B.6) is the one-dimensional generalized diffusion with effective speed $v_{\mathrm{eff}}(x) = \sqrt{2(E - E_{n(x)} - V_1(x))}$, classified by Lemma 8.2 with the* effective time of flight $T_{\mathrm{eff}} = \int_{x_*}^{\infty} \mathrm{d}x/v_{\mathrm{eff}}$.

Proof. Scale-and-speed constructions need only a strictly increasing continuous scale and a Radon speed measure ([29], §4); v_{eff} is measurable, locally bounded and locally bounded away from 0 on the open region, and the regular diffusion it defines coincides with the switched process between switch points by locality. The classification integrals are as in Lemma 8.2 ($\Sigma = N = T_{\mathrm{eff}}^2$ for $\beta = 0$). □

T_{eff} is a *path functional of the coupled mechanism*, not a function of $\{T_n\}$. The next two propositions make the underdetermination rigorous in both directions.

Proposition B.9 (Counterexample I: all channels explosive, coupled conservative; proved)**.** *There exist V_1 (continuous, non-increasing to $-\infty$, and—after a slight shelf tilt and corner mollification, which do not affect the estimates—strictly decreasing and C^1, meeting the regularity of*

Theorem 8.1) and a transverse spectrum (E_n) *with* $\sup_n T_n \le \frac{3}{2}$ *(every frozen channel explosive, uniformly—(U) holds), yet a position-locked riding law gives* $T_{\text{eff}} = \infty$*: the coupled process is* conservative. *So "*$T_n < \infty\ \forall n \Rightarrow$ *explosive" is* false *for unbounded-rate transfer, even under (U); the bounded-rate hypothesis of Theorem B.6 cannot be dropped.*

Proof. Take $E = 0$, $E_n = 16^n$, and the shelf potential $V_1 = -E_n - \frac{1}{2}$ on $[2n, 2n+1]$, linear on $[2n+1, 2n+2]$ $(n \ge 1)$. Regularity: $|V_1'| \asymp 16^n$ on the n-th drop while $|V_1|^{3/2} \asymp 16^{3n/2}$, so $|V_1'||V_1|^{-3/2} \to 0$. Channel m is open from its shelf onward, $x_m = 2m$, and $T_m \le \frac{1}{\sqrt{1}} + \frac{\sqrt{1+2(E_{m+1}-E_m)}-1}{E_{m+1}-E_m} + \sum_{n>m}(\frac{1}{\sqrt{2(E_n-E_m)}} + \frac{2}{\sqrt{E_n}}) \le \frac{3}{2}$ (numerically $T_1 = 1.17$, $T_3 = 1.01$, $T_8 = 1.00001$; `scripts/appD_escape_dictionary.py`). Riding $n(x) = n$ on $[2n, 2n+2)$: on shelf n the effective speed is 1 over unit length, so $T_{\text{eff}} \ge \sum_n 1 = \infty$; Lemmas B.8 and 8.2 finish. □

Proposition B.10 (Counterexample II: all channels conservative, coupled explosive—ghost-specific; proved)**.** *Let* $V_1 \equiv 0$ *on* $[1, \infty)$*,* $E = 1$*, and* $h_\perp$ *a* ghost *oscillator,* $E_n = -(n + \frac{1}{2})$*, unbounded below. Every frozen channel has* $T_n = \infty$ *(conservative). The riding law* $n(x) = \lceil x^{2+\epsilon} \rceil$ *gives* $v_{\text{eff}}(x)^2 = 2(E + n(x) + \frac{1}{2}) \ge 2x^{2+\epsilon}$*, so* $T_{\text{eff}} \le \int^\infty x^{-1-\epsilon/2} \mathrm{d}x/\sqrt{2} < \infty$*: the coupled process is* explosive*. The mechanism extracts unbounded energy from the transverse ghost—the stochastic realization of the channel-tail escape of Sec. 10 (the* $\lambda < 0$ *PU climbing through arbitrarily high mode-2 channels).*

Proof. Immediate from Lemmas B.8 and 8.2. □

Theorem B.11 (ghosts are necessary for Counterexample II; proved under (D),(R))**.** *Suppose* $h_\perp$ *is bounded below,* $E_n \ge E_0$*, let* $\bar{v}(x)^2 = 2(E - E_0 - V_1(x))$ *dominate every channel speed, and assume* (D) : $\int^\infty \mathrm{d}x/\bar{v} = \infty$ *and* $\limsup_{x\to\infty} \bar{v}'(x) < \infty$ *(which holds if* $-V_1' \le K\sqrt{1 + |V_1|}$*, i.e.* $-V_1' \le K\bar{v}$*), and* (R)*: every channel reflects at the common base* x_* *(no channel reflects above* x_**). Then* every *switched escape process—arbitrary adapted switching with countably many switches, including position-locked riding and,* provided the switching is itself non-explosive *(finitely many switches in every finite time interval a.s.; for a birth-chain coupling,* $\sum_n 1/q_n = \infty$*), any unbounded rates—is conservative* in the spatial coordinate X *(no escape to* $x = \infty$*); escape to the channel tail* $n = \infty$ *is governed separately by Proposition B.13. In particular, at-most-quadratic escape ends with non-ghost transverse dynamics are stable under channel coupling: under* (R) *and non-explosive switching the conservative leg of the dictionary is robust, the explosive leg is not. (The Zeno rates* $q_n = 2^n$*, for which* $\sum_n 1/q_n < \infty$*, show the non-explosive-switching hypothesis cannot be dropped: the label reaches* $n = \infty$ *in finite time at bounded* x*.)*

Proof. Khasminskii test with the dominating clock (Khasminskii, *Stochastic Stability of Differential Equations*, 2nd ed., Springer 2012, Thm. 3.5). Let $\bar{\tau}(x) = \int_{x_*}^x \mathrm{d}y/\bar{v}$ and $\varphi = \bar{\tau}$ for $x \ge x_* + 1$, smoothed to constant near the base so $\varphi' \ge 0$ vanishes at every reflection point (all at x_* by (R)); $\varphi \uparrow \infty$ by (D). For any channel n open at x $(v_n \le \bar{v})$, using $v_n v_n' = -V_1' = \bar{v}\,\bar{v}'$ (shared across channels), $v_n(v_n\varphi')' = \bar{v}'(1 - v_n^2/\bar{v}^2) \le C$, so $L_n\varphi \le \beta + \frac{\varepsilon^2}{2}C =: C'$ uniformly in n, x. Since X is continuous across switches and φ' vanishes at reflection points, $\varphi(X_{t\wedge\zeta_k}) - C't$ is a supermartingale up to the exit time ζ_k of $[x_*, k]$; $\mathbb{E}\,\varphi(X_{t\wedge\zeta_k}) \le \varphi(x_0) + C't$ while $\varphi(k) \to \infty$ gives $\mathbb{P}(\zeta_k \le t) \to 0$: no explosion. □

Remark B.12 (where separability is and is not needed). *Not needed* for the per-channel statements under bounded mixing (Theorem B.6), nor for the conservative side when $h_\perp$ is bounded

below and the end at most quadratic (Theorem B.11, under (R) and non-explosive switching, no coupling hypothesis). *Needed*: some control of transfer (bounded rate, or transverse boundedness below) for the explosive side; Proposition B.9 shows $\{T_n\}$ does not determine the coupled classification under unbounded transfer, and Proposition B.10 shows the conservative side fails only by extracting unbounded transverse energy, i.e. only for ghosts (Theorem B.11). The deciding object is the time-of-flight functional T_{eff} along the actual coupled mechanism—one integral per escape *path family*, not per channel.

B.6 The second boundary: the channel tail

In $d > 1$ the state space $\{(x, n)\}$ has, besides the channel ends at $x = \infty$, a second end: $n \to \infty$ at co-moving x.

Proposition B.13 (tail end of the channel chain; proved)**.** *Model the transverse transfer as a minimal birth chain on $\mathbb{N}$ with upward rates $q_n > 0$. The tail $n = \infty$ is reached in finite time a.s. iff $\sum_n 1/q_n < \infty$, and the mean climbing time is $\sum_n 1/q_n$ (Feller's explosion criterion; [30], Ch. 14). The tail carries its own time of flight $T_{\text{tail}} = \sum_n 1/q_n$, in exact formal analogy with $T = \int \mathrm{d}x/v$*—the dictionary acquires a second leg of the same shape*, but its datum q_n comes from the coupling, not from any integral of the potential.*

Open Problem B.14 (coupled dictionary; conjecture)**.** We conjecture: for the switched escape process with adapted rates derived from a (possibly unbounded) coupling, the process explodes iff the combined time-of-flight functional is finite along some accessible escape route, over routes carrying both spatial ends (classified by T_{eff}, Lemma B.8) and the tail end (classified by T_{tail}-type sums, Proposition B.13). The results above are the proved instances; the general statement is open because accessibility of a route for the given rates is a large-deviations question we have not settled.

Remark B.15 (what this says about Open Problem 10.6). The paper's remaining gap—whether the finite-channel indices survive $M \to \infty$ (Theorem 10.8, Corollary 10.9)—is the quantum image of the dichotomy proved here. For $\lambda > 0$ PU the classical escape locks onto a definite kinetic direction (bounded channel occupation), the regime of Theorem B.6(b)/(U) where explosivity is stable—so the analysis *supports* deficiency (∞, ∞) but does not prove it (the couplings are polynomial, not bounded; the needed strengthening is the open tail control). For $\lambda < 0$ PU the escape is channel-tail riding (Proposition B.10/ Theorem B.11), where the classical and quantum legs can and apparently do split (classical blowup, quantum essential self-adjointness at every finite M): zero-point transverse energy renormalizes the rates q_n, closing the tail. The framework locates the split at the tail end and explains its mechanism, but does not decide the full-operator question (Open Problem 10.6). The direction of robustness proved here (conservative side stable, explosive side fragile) matches the paper's numerics: both completion candidates sit on the classically-explosive-but-quantum-safe side, never the reverse.

C The extension–lateral parametrization

This appendix proves Theorem 9.1 and supplies the full parametrization behind Sec. 9: the boundary unitary of a self-adjoint extension (Proposition C.1), the flux functor onto the slice $\mathcal{C}$ of the Feller–Wentzell cone, the four exclusions each tied to its exact invariant—including the maximal-dissipative matrix-ball picture—and the honest caveats (E-covariance and the $\varepsilon > 0$

jump-rate normalization). Status conventions as in Appendix B: statements are *[proved]* unless tagged *[proved, sketch]* or *[conjecture]*; everything at the class level is *[proved]*, and the only WKB input the class-level algebra uses is the Olver-normalized in/out frame of §C.1 (which fixes the SU(1, 1) frame freedom so that the moduli $|S_{ij}|^2$ are well-defined); the finer dynamical asymptotics enter only in Proposition C.4.

C.1 Boundary data on the two sides

Throughout, $\tau\psi = -\frac{1}{2}\psi'' + V\psi$ on $I = (r_-, r_+)$ with $r_+ = +\infty$ an escaping end of the mechanism model of Theorem 8.1 in its limit-circle regime ($V \in C^2$, $V \to -\infty$ eventually monotonically, $|V'|\,|V|^{-3/2}$ bounded and $|V''|\,|V|^{-3/2}$ integrable—the added C^2 control of leg (2)) and classical time of flight $T_+(E) = \int^\infty \mathrm{d}x/\sqrt{2(E-V)} < \infty$, so that r_+ is limit circle for τ [23] and a *regular* Feller boundary for the escape diffusion (Lemma 8.2(i)). Two cases: $n = 1$ (r_- confining, hence limit point with no boundary data; deficiency $(1, 1)$, extensions $\cong U(1)$) and $n = 2$ ($r_- = -\infty$ also escaping with $T_-(E) < \infty$, e.g. $V = -x^4$; deficiency $(2, 2)$, extensions $\cong U(2)$). Fix once and for all a real E with $V < E$ on a neighborhood of each escaping end—the same E at which the escape diffusion $\mathrm{d}X = \sqrt{2(E - V(X))} \circ (\mathrm{d}t + \varepsilon\,\mathrm{d}W)$ of Lemma 8.2 (or its driftless variant, scale $s = \tau_{\mathrm{clock}}$, speed $m = 2\,\mathrm{d}x/v$) is built. For $n = 1$ the confining end plays the role of the reflecting base point x_0; for $n = 2$ both ends carry boundary data.

Quantum boundary data *[proved].* For $\psi, \varphi \in D(\tau_{\max})$ the Lagrange bracket $[\bar\psi, \varphi]_x = \bar\psi\varphi' - \bar\psi'\varphi$ has limits at each end, and Green's identity localizes the boundary form there:

$$\langle \tau\psi, \varphi\rangle - \langle \psi, \tau\varphi\rangle = \tfrac{1}{2}\Big([\bar\psi, \varphi]_{r_+} - [\bar\psi, \varphi]_{r_-}\Big) =: \mathrm{i}\,\Omega(\psi, \varphi). \tag{18}$$

Ω here (the boundary current form, distinct from the vacuum Ω of Sec. 6 and the squared-frequency sum $\Omega = \omega_1^2 + \omega_2^2$ of Sec. 10) descends to the $2n$-complex-dimensional boundary space $\mathcal{B} = D(\tau_{\max})/D(\tau_{\min})$ (with τ the Sturm–Liouville expression, not the rescaled clock $\tau = \kappa t$ of Sec. 5 or the Feller clock coordinate of Lemma 8.2), where it is a nondegenerate Hermitian form of signature (n, n); self-adjoint extensions are in bijection with the *maximal* Ω*-neutral* subspaces of $\mathcal{B}$ (complex-symplectic/boundary-triple theory [32, 33, 31]). Physically, $\Omega(\psi, \psi)$ is the total *outward probability current* at the ends: with $j(\psi) = \mathrm{Im}(\bar\psi\psi')$ one has $\partial_t|\psi_t|^2 + \partial_x j(\psi_t) = 0$ pointwise in I for *every* extension (all extensions act as the same differential expression), and $\Omega(\psi, \psi) = j(\psi)(r_+) - j(\psi)(r_-)$.

At each escaping end, the solution germs $N_E^\pm = \{u : \tau u = Eu \text{ near } r_\pm\}$ form a 2-complex-dimensional space on which the per-end current form

$$J^{r+}(u, w) = \tfrac{1}{2\mathrm{i}}\,[\bar u, w]_{r_+}, \qquad J^{r-}(u, w) = -\tfrac{1}{2\mathrm{i}}\,[\bar u, w]_{r_-} \quad \text{(sign: current } \textit{toward} \text{ the end)}$$

is constant in x, Hermitian, of signature $(1, 1)$. A real pair u_1, u_2 with Wronskian 1 gives combinations $\psi_\pm = u_1 \pm \mathrm{i}u_2$ with $J(\psi_\pm, \psi_\pm) = \pm 1$, $J(\psi_+, \psi_-) = 0$; but the $\mathrm{SL}(2, \mathbb{R})$ freedom in that pair induces an end-local SU(1, 1) freedom in (ψ_+, ψ_-), under which the scattering moduli $|S_{ij}|^2$ below are *not* invariant, so an arbitrary real pair does not by itself fix $R = |S_{ii}|^2$. We therefore *pin* the frame to the physical in/out waves: under the mechanism hypotheses (with $V \in C^2$ and $|V''|\,|V|^{-3/2}$ integrable, as in Prop. C.4) the Liouville–Green solutions $\psi_\pm \sim v^{-1/2}\mathrm{e}^{\pm\mathrm{i}\int v\,\mathrm{d}x}$, $v = \sqrt{2(E-V)}$, exist with Olver's error bounds [35] and are canonical up to a common $U(1)$ phase (their real and imaginary parts form a Wronskian-1 real pair), and we take $\psi_\pm$ to be these. Under the residual diagonal $U(1)$ the moduli $|S_{ij}|^2$ *are* invariant, so $R = |S_{ii}|^2$

and the flux functor below are well-defined; this Olver-normalized frame is the only WKB input the algebra requires, and the class-level statements of Theorem 9.1 (unitarity of S, double stochasticity, the $n = 1$ total reflection) are frame-independent regardless. Write $u = a\,\psi_+ + b\,\psi_-$ near each escaping end: a is the current-normalized *incident* (toward-the-boundary) amplitude, b the *returned* one, and $J(u,u) = |a|^2 - |b|^2$. Because $T_\pm < \infty$ (limit circle), the trace map $\mathrm{tr}_\partial : N_E := \bigoplus_i N_E^{(i)} \to \mathcal{B}$ (germs to boundary data) is an isomorphism for *every* $E \in \mathbb{R}$ [31].

Stochastic boundary data *[proved]*; Feller–Wentzell. At a regular boundary of the escape diffusion, with generator $G = D_m D_s$ as in Appendix A, the sub-Markov C_0 extensions of the minimal (absorbed) semigroup are in bijection with *lateral conditions*

$$p_0\,u(r_+) \;-\; p_1\,\frac{\mathrm{d}u}{\mathrm{d}s}(r_+) \;+\; p_2\,(Gu)(r_+) \;=\; \int \big(u(y) - u(r_+)\big)\,\nu(\mathrm{d}y), \tag{19}$$

$p_0, p_1, p_2 \geq 0$, $\nu \geq 0$ a measure on $\bar{I}\setminus\{r_+\}$ (the interior and, in the two-ended problem, the other boundary point), not all data zero, modulo positive scaling [27, 28, 29]. Terminology: p_0 *killing* (absorbing $= p_0$ alone, i.e. $u(r_+) = 0$, the minimal explosive semigroup; elastic $= p_0, p_1 > 0$), p_1 *reflection* (the scale derivative $\mathrm{d}u/\mathrm{d}s$ in (19) is taken into the interval—inward-pointing at the right end r_+—so the datum carries the standard endpoint-symmetric Feller–Wentzell signs; for $p_2 = \nu = 0$ the elastic condition is then $\mathrm{d}u/\mathrm{d}s = (p_0/p_1)u \geq 0$ inward, the sub-Markov sign), p_2 *stickiness* (the boundary carries a speed-measure atom $p_2\delta_{r_+}$: the process spends positive Lebesgue time at r_+), ν *jump-in* (re-entry, possibly at the other boundary: Feller's "circle-type" passage). Applying (19) to $u = \mathbf{1}$ shows: *the extension is conservative iff $p_0 = 0$ at every regular end* (no killing anywhere; ν may be arbitrary). Denote by $\mathcal{W}$ the set of lateral data (one datum per regular end), a cone of infinite dimension (because of ν).

C.2 The boundary unitary of a completion

Proposition C.1 (boundary scattering matrix at fixed energy; proved)**.** *Fix E as above and let $\tilde{H}$ be any self-adjoint extension, with maximal Ω-neutral boundary subspace $L \subset \mathcal{B}$. Let $\mathcal{S}_L = \{u \in N_E : \mathrm{tr}_\partial\, u \in L\}$ (germ solutions at energy E admissible for $\tilde{H}$), and let $a = (a_i)$, $b = (b_i) \in \mathbb{C}^n$ be the incident/returned amplitude coordinates at the escaping ends. Then:*

(i) $\dim_{\mathbb{C}} \mathcal{S}_L = n$, *and for* $u \in \mathcal{S}_L$, $\sum_i \big(|a_i|^2 - |b_i|^2\big) = 0$ (exact current conservation)*;*

(ii) the map $u \mapsto a$ is injective on $\mathcal{S}_L$, hence bijective onto $\mathbb{C}^n$, and there is a unique $S_{\tilde{H}}(E) \in U(n)$ with $b = S_{\tilde{H}}(E)\,a$ on $\mathcal{S}_L$;

(iii) $\tilde{H} \mapsto S_{\tilde{H}}(E)$ is a bijection from the self-adjoint extensions onto $U(n)$. Separated boundary conditions correspond to diagonal S (the torus $U(1)^n \subset U(n)$); for $n = 1$, $S_\vartheta(E) = \mathrm{e}^{\mathrm{i}\alpha(\vartheta,E)}$ recovers the $U(1)$ of Theorem 8.1(2)(a).

For comparison, the same construction with "maximal neutral" relaxed to "maximal nonnegative" ($\Omega(u,u) \geq 0$ on L, i.e. net outward current ≥ 0: maximal-dissipative extensions, contraction semigroups $\mathrm{e}^{-\mathrm{i}t\tilde{H}}$) yields exactly the closed matrix ball $\|S\| \leq 1$ [33]; the self-adjoint family is its unitary boundary.

Proof. (i) The trace map $N_E \to \mathcal{B}$ is an isomorphism (limit circle at each escaping end; the confining end contributes no data), so $\mathcal{S}_L \cong L$, of dimension n. For $u \in \mathcal{S}_L$: $\Omega(u,u) = 0$ by

neutrality of L, and $\Omega(u,u) = \sum_i J^{(i)}(u,u) = \sum_i(|a_i|^2 - |b_i|^2)$ by constancy of the per-end bracket on solutions.

(ii) If $a = 0$ then (i) gives $\sum |b_i|^2 = 0$, so $u = 0$: injective; by dimension count, onto $\mathbb{C}^n$. Define S by $b = Sa$; (i) says $\|Sa\| = \|a\|$ for all a, i.e. $S \in U(n)$.

(iii) $\mathcal{S}_L$ is recovered from S as $\{a\,\psi_+ + Sa\,\psi_-\}$ and $L = \mathrm{tr}_\partial\,\mathcal{S}_L$; conversely, any maximal neutral subspace of a signature-(n,n) space is the graph of a unitary from the positive to the negative part (if $(0,b') \in L$, neutrality forces $b' = 0$; then the a-projection is bijective and the graph map is isometric). In the (a,b) coordinates at energy E, $\Omega = \mathrm{diag}(+I_n, -I_n)$, so maximal neutral subspaces $\leftrightarrow U(n)$, and by the trace isomorphism these are exactly the extension subspaces. Separated conditions constrain each end's data independently, giving block-diagonal (here 1×1, i.e. diagonal) S. For the dissipative case replace "isometric graph" by "contractive graph." □

Remark C.2 (E-covariance). The extension $\tilde{H}$ is fixed; the frames $\psi_\pm(E)$ rotate with E by an end-local $U(1,1)$ Möbius action, under which unitarity of $S(E)$ is preserved but the moduli $|S_{ij}(E)|$ in general drift with E. Two subfamilies are E-uniform: separated conditions (diagonal S for all E) and the smooth gluing (the periodic identification of Fig. 7: the glued clock-circle problem has no defect at the seam, so S is antidiagonal—perfect transmission—at every E). The dictionary below is stated at the mechanism energy E, the same E at which the escape diffusion itself is defined; this matching of reference energies is intrinsic to the construction, not a defect.

C.3 The flux functor and the proof of Theorem 9.1

Definition C.3 (flux functor)**.** For a self-adjoint extension $\tilde{H}$ define the lateral datum $\Phi_E(\tilde{H}) \in \mathcal{W}$ by: at each escaping end i,

$$p_0^{(i)} = 0, \qquad p_2^{(i)} = 0, \qquad p_1^{(i)} = |S_{ii}(E)|^2, \qquad \nu^{(i)} = \sum_{j\neq i} |S_{ij}(E)|^2\,\delta_{r_j},$$

i.e. conservative, instantaneous, boundary-supported, with reflection weight $|S_{ii}|^2$ and passage-to-the-other-end weight $|S_{ij}|^2$. (For $\varepsilon > 0$ the pair (p_1,ν) is to be read projectively—re-entry at local-time rate ν/p_1; in the transport limit $\varepsilon \downarrow 0$ the weights are literal per-arrival branching probabilities. See Remark C.5.)

Proof of Theorem 9.1. (1)–(3). By Proposition C.1, $\tilde{H} \mapsto S(E)$ is a bijection onto $U(n)$, and Φ_E factors through it. For $n = 1$, $|S| = 1$ forces reflection weight 1: the constant map, with fiber the whole $U(1)$. (Equivalently, without Proposition C.1: any separated real boundary condition $\cos\vartheta\ \mathrm{tr}_{\partial,1}\,u + \sin\vartheta\ \mathrm{tr}_{\partial,2}\,u = 0$ is real, so its admissible germ line is spanned by a *real* solution $u = a\psi_+ + \bar{a}\psi_-$, whence $|b/a| = 1$: total reflection for every ϑ.) For $n = 2$, double stochasticity of $(|S_{ij}|^2)$ (row and column sums 1, because S is unitary) gives $|S_{11}|^2 = |S_{22}|^2 =: R$ and $|S_{12}|^2 = |S_{21}|^2 = 1 - R$, so the image lies in the reciprocal family of part (3), which is contained in $\mathcal{C}$ by construction of Φ_E. Ontoness: the matrix $S = \left(\begin{smallmatrix} \sqrt{R} & \mathrm{i}\sqrt{1-R} \\ \mathrm{i}\sqrt{1-R} & \sqrt{R} \end{smallmatrix}\right)$ is unitary for every $R \in [0,1]$; $R = 1$ is realized by the separated (e.g. Dirichlet) completions and $R = 0$ by the smooth gluing, whose stochastic shadow is Feller's circle-type jump datum—the darning of the two boundary points [34]. Fibers: fixing the moduli of a 2×2 unitary leaves its four phases free up to one orthogonality relation (columns orthogonal), a torus of dimension 3.

(4): the four exclusions, each by its exact invariant.

(c1) Killing ($p_0 > 0$: absorbing, elastic) is excluded by unitarity. If $p_0 > 0$, start the process at any interior x; since the boundary is regular and accessible ($\Sigma < \infty$, Lemma 8.2(i)–(ii)), $\mathbb{P}_x(\text{hit the end} \le t) > 0$ for every $t > 0$, and each boundary visit kills a positive fraction of local-time mass, so $\mathbb{E}_x[\mathbf{1}_{\{\text{alive}\}}] < 1$ for all $t > 0$; under any correspondence matching total probability this would equal $\left\|\mathrm{e}^{-\mathrm{i}t\tilde{H}}\psi\right\|^2 / \|\psi\|^2$, which a self-adjoint (norm-preserving) evolution holds at 1—so no self-adjoint extension matches a killing datum. The dissipative match is Proposition C.1: maximal nonnegative boundary subspaces are contractive graphs $\|S\| \le 1$; $S = 0$ is the condition $b = 0$ (no returned amplitude: every admissible germ purely outgoing), the quantum radiation condition, whose semigroup loses exactly the arriving flux, as does the absorbing/minimal explosive semigroup of Theorem 8.1(2)(c); $0 < \|S\| < 1$ returns the fraction $\|Sa\|^2 / \|a\|^2 < 1$, as does the elastic condition (per-arrival kill fraction $1 - |S|^2$ in the transport limit). Hence sub-unitary $\leftrightarrow$ sub-Markov, and the conservative slice is matched by exactly the extreme points (the unitaries) of the ball.

(c2) Stickiness ($p_2 > 0$) is excluded by the measure class of the state space. For a sticky datum, $\mathbb{E}\int_0^t \mathbf{1}_{\{X_s = r_+\}}\mathrm{d}s > 0$ after the first boundary hit (speed atom $p_2\delta_{r_+}$; [29], §3.7), so $\mathbb{P}(X_s = r_+) > 0$ on a time set of positive measure, and the semigroup is symmetric only in $L^2(m + p_2\delta_{r_+})$; every self-adjoint extension instead acts in $L^2(I, \mathrm{d}x)$, whose measure has no boundary atom: $|\psi_t|^2\mathrm{d}x(\{r_+\}) = 0$ for all t. A quantum counterpart would require enlarging the Hilbert space to $L^2(I) \oplus \mathbb{C}$ (a boundary state)—extensions of a *different* symmetric operator, outside the $U(n)$ family (Remark C.7).

(c3) Interior jump-in ($\nu(I^\circ) > 0$) is excluded at the classification level. By (18) the boundary form of τ vanishes on compactly supported interior functions: the self-adjointness constraint is a condition on *end* data alone, and the classification of extensions by maximal neutral subspaces of the end boundary space is exhaustive, so no extension has a parameter available to encode an interior re-entry point; Φ_E's jump measures are supported on the ends by construction, and no refinement through boundary data can do otherwise. (Semiclassically the same: wave packets return through the ends.)

(c4) Asymmetric passage is excluded by unitarity again. For $U \in U(2)$ the matrix $(|U_{ij}|^2)$ is doubly stochastic, hence $|S_{12}|^2 = |S_{21}|^2$ and $|S_{11}|^2 = |S_{22}|^2$. Independent lateral data at the two ends form a two-parameter family (R_+, R_-); the image is exactly its diagonal. □

C.4 Dynamical content *[proved, sketch]*

Proposition C.4 (arrival and revival; proved, sketch)**.** *For a WKB wave packet of mean energy E launched toward end i: it reaches the boundary after the classical time of flight ($\approx T_i(E)$, the $\varepsilon \downarrow 0$ explosion time of the escape diffusion, Lemma 8.2(ii)), and the fraction $|S_{ji}(E)|^2$ of its probability re-emerges from end j after the finite round trip. This matches the $\varepsilon \downarrow 0$ escape diffusion with branching lateral datum $\Phi_E(\tilde{H})$, for which the boundary behavior per arrival is literally "reflect w.p. R / pass w.p. $1 - R$." The revival numerics of Fig. 7 (right)—Dirichlet wall returning from the right, periodic gluing from the left—are the two extreme points $R = 1, 0$.*

Proof sketch (the only non-exact leg). Under the mechanism hypotheses, augmented by $V \in C^2$ with $|V''|\,|V|^{-3/2}$ integrable (which implies, but is not implied by, $V'(-V)^{-3/2}$ of bounded variation), the Liouville–Green solutions carry Olver-type error bounds ($\int |v^{-1/2}(v^{-1/2})''|\,\mathrm{d}x < \infty$, whose integrand $\frac{5}{16}q^{-5/2}(q')^2 - \frac{1}{4}q^{-3/2}q''$ with $q = 2(E - V)$ has its V'' piece controlled by the $|V''|\,|V|^{-3/2}$ integrability and its $(V')^2$ piece by the same, through $\int (W')^2 W^{-5/2} = -\frac{2}{3}\left[W'W^{-3/2}\right] + \frac{2}{3}\int W''W^{-3/2}$ ($W = E - V$), the boundary term $|V'|\,|V|^{-3/2}$ bounded with

$V \to -\infty$ monotonically), so $\psi_\pm = v^{-1/2}\mathrm{e}^{\pm\mathrm{i}\int v}(1+o(1))$ with controlled remainders [35]. A packet $\int \chi(E')\,u_{E'}\,\mathrm{e}^{-\mathrm{i}E't}\,\mathrm{d}E'$ built on admissible solutions ($b(E') = S(E')a(E')$) has, by stationary phase, an incident front arriving at the end at the classical time and returned fronts of weight $|S_{ji}|^2$ emerging at end j, with corrections $O(\text{window}^{-1}) + o(1)_{\text{WKB}}$; the discrete spectrum (spacing π/T_{cross}, Theorem 8.1(2)(b)) does not obstruct because the packet width covers many levels. On the stochastic side, $\varepsilon \downarrow 0$ reduces the escape diffusion to the transport flow $\dot{x} = v$ (Lemma 8.2(ii), drifted variant: $\zeta \to T$ a.s.), for which the branching lateral datum acts per arrival with probabilities $R, 1-R$. The two evolutions then agree at the level of arrival times and returned/passed mass fractions. This leg is asymptotic by nature; the exact statements are Theorem 9.1(1)–(4). □

C.5 Caveats and the collapse mechanism

Remark C.5 (rate normalization at $\varepsilon > 0$). At $\varepsilon > 0$ a regular boundary is visited on a local-time set, so "reflect w.p. R per bounce" is ill-posed; the correct ν-versus-p_1 normalization is an excursion-theoretic convention (jump at local-time rate $\propto (1-R)/R$), and only the $\varepsilon \downarrow 0$ limit makes the weights literal per-arrival probabilities. We therefore claim the quantitative identification of R only in that limit; the class-level statements (which slice of the cone, which ends linked, reciprocity) are exact for every $\varepsilon > 0$.

Remark C.6 (why the phases must collapse). The collapse of the $U(1)$ fibers is structural, not an artifact of the map chosen. On the escaping end $v \geq v_{\min} > 0$, so the quantum phase $\int^\infty v\,\mathrm{d}x = \infty$ diverges while the clock $\int^\infty \mathrm{d}x/v = T < \infty$ converges: the quantum boundary basis $\psi_\pm$ winds *infinitely often* approaching the end (solutions have infinitely many zeros accumulating at $+\infty$), whereas the diffusion's boundary basis—the G-harmonic functions 1 and s—is monotone with finite boundary limits. No linear identification of the two two-dimensional boundary spaces can be canonical (there is no preferred "zeroth winding"), and—once the frame is pinned to the Olver in/out waves of §C.1, which fixes the $\mathrm{SU}(1,1)$ freedom up to the residual diagonal $U(1)$—the only quantities invariant under that residual and transferable to the probabilistic side are the current moduli $|S_{ij}|^2$. The phase is precisely the datum the $|\cdot|^2$ functor forgets.

Remark C.7 (what would see the phases; conjecture). Two natural extensions of the stochastic side, both outside Feller's lateral-condition cone for the escape diffusion in $L^2(m)$, are the plausible receptacles for the lost data, and we record them as *conjectural* dictionary entries: (a) sticky data $p_2 > 0$ correspond to self-adjoint operators on $L^2(I) \oplus \mathbb{C}$ (a boundary degree of freedom; Wentzell-type quantum boundary conditions), not to any extension of H in $L^2(I)$; (b) the Wigner delay $\partial\alpha/\partial E$ of a fixed extension is the natural phase shadow of boundary delay, but it can be negative and is not constant-per-visit, so it does *not* match any sub-Markov sticky parameter—consistent with (c2) being an exact exclusion rather than a near miss.

C.6 The correspondence table

Table 3 collects the dictionary. In summary: the class-level dictionary—the image characterization, the $n = 1$ collapse, the $n = 2$ segment, all four exclusions, and sub-unitary $\leftrightarrow$ sub-Markov—is *proved*; only the wave-packet arrival/revival interpretation is semiclassical (Proposition C.4), and the $\varepsilon > 0$ jump-rate normalization is conventional (Remark C.5). Theorem 8.1(2)(a)–(b) and Lemma 8.2(iii) are correct as worded ("stochastic mirror," "parametrize"), provided the mirror is not sharpened to a bijection; the sharp form is: *the conservative, instantaneous,*

Quantum completion (at E)	**Flux data $S(E)$**	**Feller–Wentzell lateral datum**	**Status**
any self-adjoint ext., one end (every phase ϑ)	$e^{i\alpha}$, $\lvert S\rvert = 1$	reflecting: $\mathrm{d}u/\mathrm{d}s(r_+) = 0$	proved (class); timing WKB
separated completions, two ends $(\vartheta_+, \vartheta_-)$	diagonal, $R = 1$	independent reflection at each end	proved
smooth gluing ("periodic through ∞")	antidiagonal, $R = 0$, all E	passage $r_+ \leftrightarrow r_-$ (circle-type jump; darning)	proved (class)
generic coupled $U \in U(2)$	$0 < R < 1$	reflect wt. R + jump to other end wt. $1 - R$, same at both ends	proved (class); R generically E-dependent; rate conv. Rem. C.5
relative phases $\arg S_{ij}$	invisible	— (fibers of Φ_E; eigenvalue positions)	proved
maximal dissipative, radiation cond. $b = 0$	$S = 0$	absorbing $u(r_+) = 0$: minimal explosive semigroup	proved (flux); *not* self-adjoint
strictly contractive boundary data	$0 < \lVert S\rVert < 1$	elastic ($p_0, p_1 > 0$: partial kill)	proved (flux); *not* self-adjoint
none in $L^2(I, \mathrm{d}x)$ (needs $L^2 \oplus \mathbb{C}$)	—	sticky ($p_2 > 0$)	exclusion proved; enlarged-space entry conjectural
none (no interior channels among the extensions)	—	jump-in to interior ($\nu(I^\circ) > 0$)	exclusion proved
none (double stochasticity)	—	asymmetric passage $R_+ \neq R_-$	exclusion proved

Table 3: The extension–lateral dictionary (Theorem 9.1) at the mechanism energy E. Top block: the self-adjoint family and its flux shadows (the slice $\mathcal{C}$). Middle block: the sub-Markov data matched by maximal-*dissipative*, not self-adjoint, completions. Bottom block: lateral data with no counterpart at all, each excluded by an exact invariant. The Dirichlet-wall and periodic-gluing revivals of Fig. 7 are the extreme points $R = 1$ and $R = 0$.

boundary-supported, reciprocal lateral conditions are exactly the flux shadows of the self-adjoint family, the phases being spectral data with no stochastic counterpart.

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